

Open Channel UV System

The Atlantium low-pressure UV array for
water disinfection

User Manual

PW0000001

September 2024



IMPORTANT NOTICE
It is strictly forbidden to alter/change the system hardware in any way.
Any change made without written permission from Atlantium shall void the warranty.



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Validations and certifications completed:



Revision History

Revision #	Description	Date
001	Initial Version	Jul 2022
002	Changed control cabinet, system diagram control and setup	Aug 2023

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1 Introduction Open Channel UV

Congratulations on purchasing an Atlantium **Open Channel UV** Hydro-Optic UV Water Treatment System. This user guide provides you with clear guidelines for preparing the site for installation, correct installation, software configuration procedures, and a good understanding of all system functions for maximum benefit from the system.

Atlantium Technologies was established in 2003 to provide the world with safe and sustainable water treatment solutions, delivering water bio-security, peace of mind and the highest quality water to our partners. Atlantium has ensured the water curiosity needs of approximately 4,500+ installations globally.

1.1 Introduction

The **Atlantium Open Channel UV Water Treatment System (OCUVS)** disinfect open channel water utilizing UV lamps cassettes that can be installed in different configurations. The water flows to the open channel through an optional Flow Meter and a sluice gate that control the flow of water through UV lamps cassettes of the OCUVS thus controlling the water level to not expose the UV lamps or exceed maximum water height.

Atlantium's different HOD-UV models systems are designed to provide applications for the following Industries:

- | | |
|--------------------|-----------------|
| ■ Aquaculture | ■ Bio-Pharma |
| ■ Fish Hatcheries | ■ Bottled Water |
| ■ Dairy | ■ Beverages |
| ■ Municipal | ■ Breweries |
| ■ Power Generation | ■ Aquariums |
| ■ Ultra Pure Water | ■ ...and more |
| ■ Marine | |

Patented Innovative Technology

- Modular design for different flow rates and water condition
- Low pressure - High Intensity based system
- Sustainable operation provides the security and reliability that the industry demands.
- Some enclosures/components as noted on the Atlantium main website under <https://atlantium.com/about.html>.
- Dose control mode or Power mode
- Optional weir with level / flow adjustment

Controlled Dose Delivery

- Monitoring & control software ensures that required UV dose is delivered at all times.
- Dedicated UV Intensity Sensor (UVIS) for each lamp for real-time monitoring.
- Automatically adjusts lamp power based on real-time analyses of critical parameters such as water flow rate, lamp intensity and water Transmission to UV (UVT).
- A user friendly interface displays the monitored data in real-time including the actual delivered dose as well as the other critical parameters, so you can verify that you are actually getting the dose that you need.

Lower Electrical Consumption

- Optimized light distribution, through Hydraulic and Fiber optic principles applied to the UV system design, enable:
- Optimized UV power and uniform dose distribution.
- Efficient use of power, with lower electrical consumption than low-pressure based UV systems.

Minimum Down Time

- True in-line system.
- Quick lamp replacement and better safety.

Integrates with your Process Line

- Field proven installation.
- With no additional software, you can integrate the Atlantium system into your current process using Atlantium's customizable settings.
- Automatically generated reports at the push of a button for performance recording and traceability.

Efficiency and Easy Maintenance

- For maximum efficiency and dependable performance, the Atlantium system monitors the lamps for UV output, ensuring that a lamp is replaced only when absolutely needed.
- By following the easy preventive maintenance programs, you keep the system working at peak performance.
- The **View port** allows you to safely peek at what's going on inside.

For further information about Atlantium products and technologies, visit our website at <https://atlantium.com/>

1.2 Before You Begin

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- You are responsible if the product is used other than its designated purpose or disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product must be used for its design purpose, based on its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

1.3 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers



DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.



WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.



ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.



Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.



Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.



Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving caustic chemicals which, if not avoided, can result in death or serious damage.



Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.



Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.



Indicates information related to installation, safety or use of system.

2 About the Open Channel UV System

This user manual describes the The Atlantium **Open Channel UV System**, which is designed for the disinfection of water.

The submittal package includes general arrangement drawing for assisting the engineer of the specific site for designing the UV setup. Please note that this drawing is general one and does not refer to a specific project. Consult with Atlantium application team for some specific guidelines for your specific project.

2.1 Open Channel UV System's Main Components

- Each **Open Channel UV System** unit consists of an array of 11 vertical low-pressure UV lamps (just 6 lamps may be required, for certain applications) which are enclosed in a steel cassette (3) that is lowered into a 750 mm-wide concrete-lined water channel (Figure 2-2 on page 8).
- A double-width configuration, featuring 2 cassettes aligned side-by-side with a total of 22 lamps, is also available for a 1,550 mm-wide channel and a t 2,350 mm-wide channel with three cassettes aligned side-by-side (Figure 2-3 on page 9).
- Depending on customer requirements and design specs, different configuration may be implemented, for example, two or more single cassettes (11 lamp) or double cassettes (22 lamp) may be arranged in series, as in Figure 2-14 on page 21.
- Each lamp is protected from the water by a quartz sleeve (1).
The lamps are powered from Lamp drivers enclosed in a nearby Lamp Drivers Electrical Cabinet (Figure 2-4 on page 12).
- An optional sluice gate (which is referred to as a “weir” in diagrams and in the software HMI - Figure 2-3 on page 9) may be operated manually or via the OCVUVS control system.
- A control cabinet (Figure 2-5 on page 12) monitors the UV output of the system via the sensors and enables automatic control of the power levels of the lamps
- A Drivers cabinet that powers the lamps. Each driver powers two lamps.

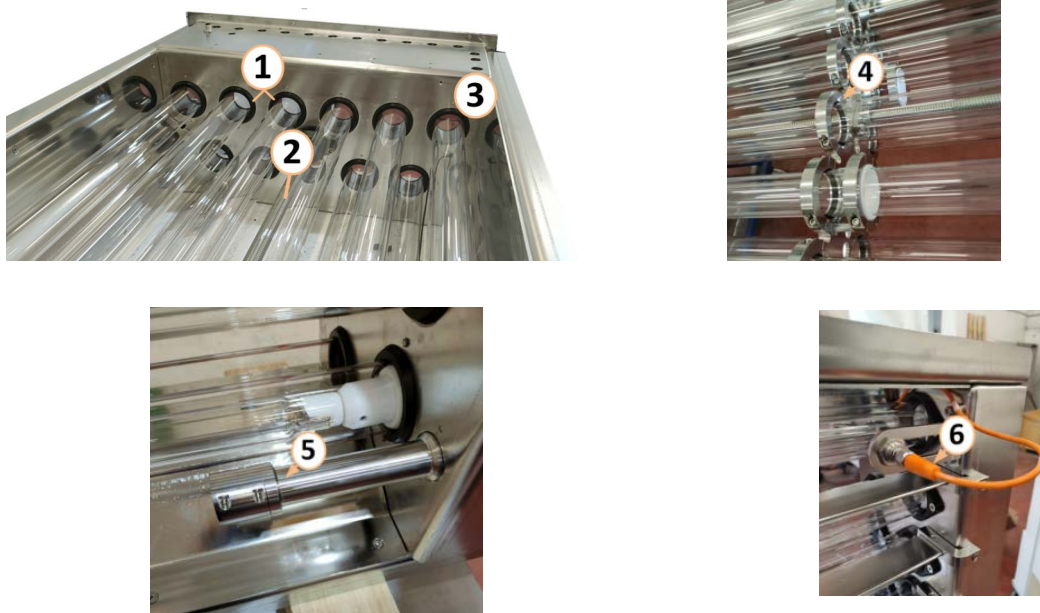


Figure 2-1: From R L to L – Top: general view of unit; ring wipers of DPM bottom: UVIS; UVTS

- One of the lamps in the system is fitted with a UV Intensity Sensor (UVIS) enclosed within the quartz sleeve. The purpose of the UVIS is to monitor the intensity of UV produced by the lamp
- An additional sensor (5), designated UV Transmission Sensor (UVTS), is positioned in the water to measure its transparency to UV light.
- The UV lamps are arranged in the unit to optimize the exposure of all the water passing through the unit to UV light, so neutralizing the bacteria in the water.
- A Deposit Prevention Mechanism (DPM), consisting of a set of ring wipers (4) moving up and down a threaded rod (2) to clean the external surfaces of the quartz tubes, helps to keep them clear of deposit.
- A pair of level sensors (6) is triggered when the set of DPM ring wipers reaches the upper / lower limits of its travel.

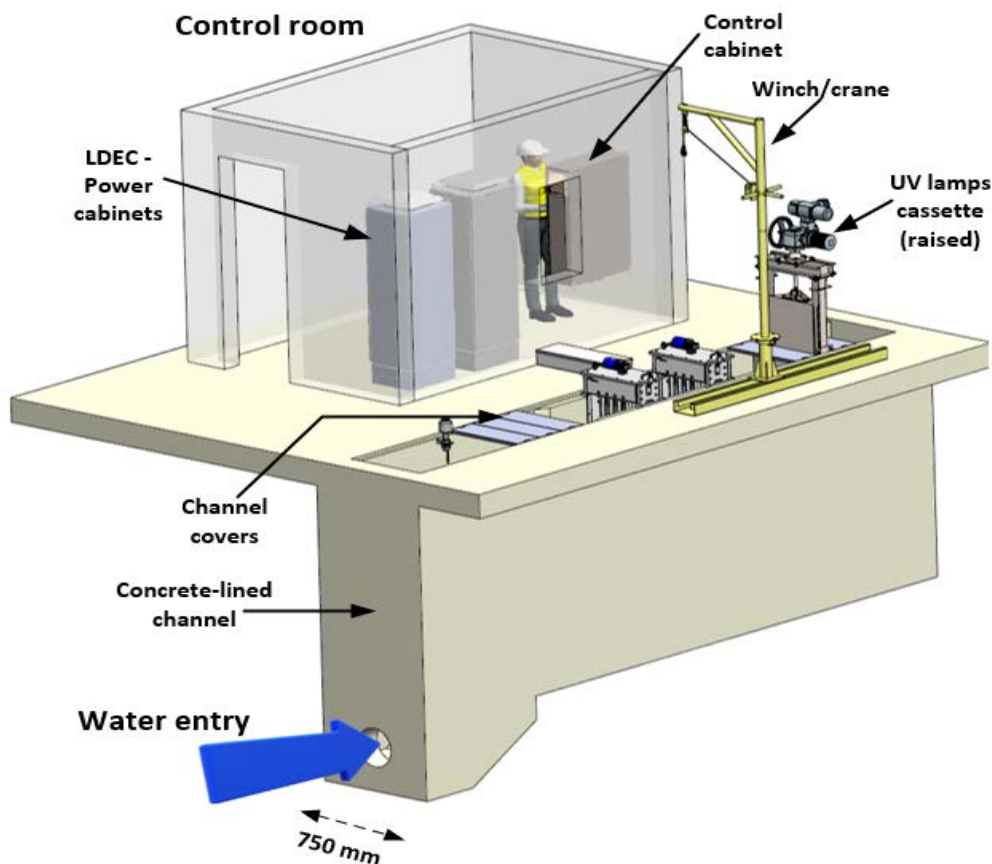


Figure 2-2: Open Channel UVS – generalized set-up with three rows of 750-mm cassettes



Please note that the system designs shown in Figure 2-2, Figure 2-3 and Figure 2-4, like other figures in this manual, are **generalized** and do not refer to your specific project. For the precise details of your system, consult the accompanying **Interface Control Drawings (ICD)**.

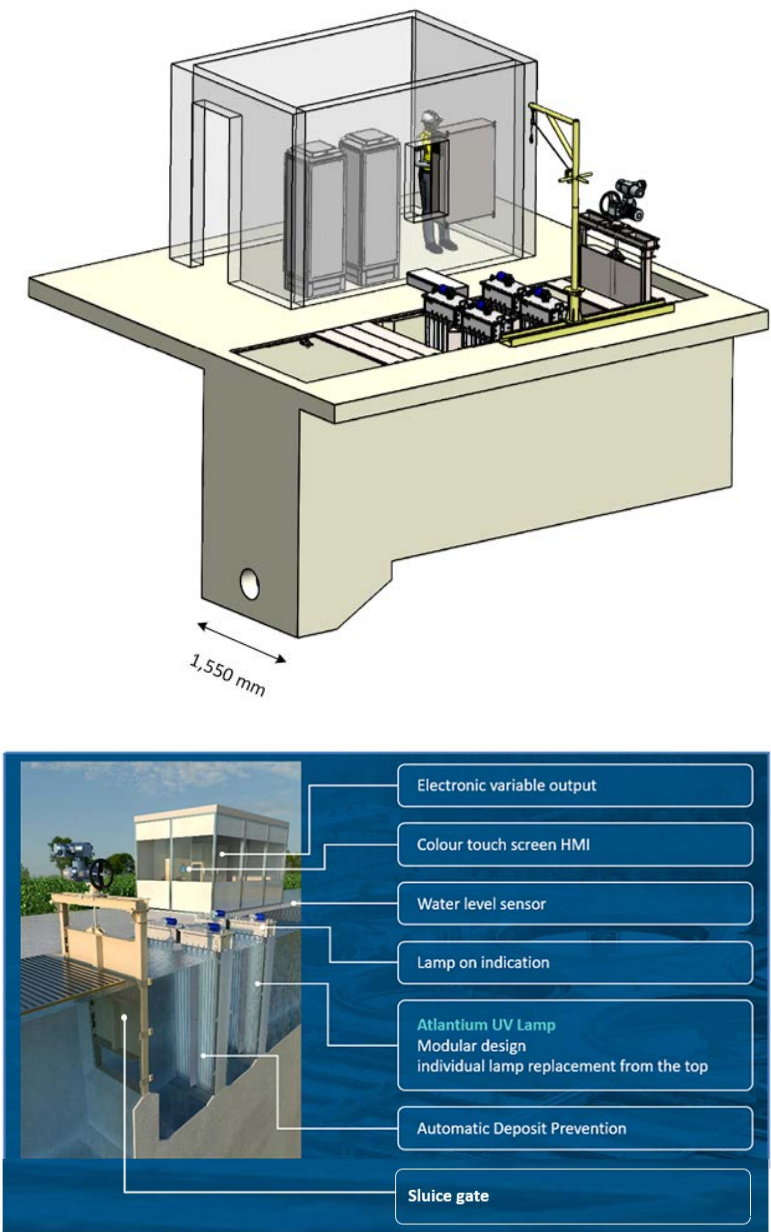


Figure 2-3: OCUVS – generalized set-up with 2 rows of double-width cassettes. Below, overall system view

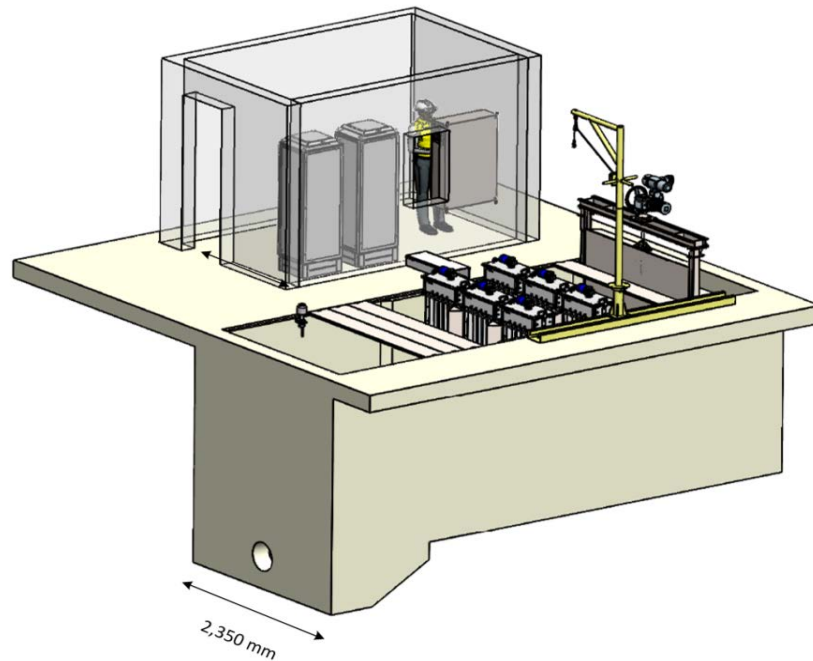


Figure 2-4: OCUVS – generalized set-up with 2 rows of triple-width cassettes

2.1.1 The OCUVS Cassette

The OCUSV unit includes 11 low pressure UV lamps that are housed within 11 quartz sleeves arranged in a vertical steel cassette (see Table 2-1 on page 11).

For certain applications, just 6 lamps are required per cassette and the other 5 sleeves remain empty.

Table 2-1: OCUVS cassette - main parts

1. Steel reactor frame assy.
2. Quartz sleeve
3. UV lamp 1.0kw, LPHI
4. Lamp conn housing assy. with UVIS
5. Lamp conn housing assy.
6. Nut lamp holder
7. UVTS assy.
8. Vane assy.
9. Spring washer, m6, DIN 7980, A4
10. Socket head cap screw, m6x16, DIN 912, A4
11. Top cover assy.
12. Connector, 5 pole, female
13. Atlantium logo
14. Bridge front
15. Bridge rear

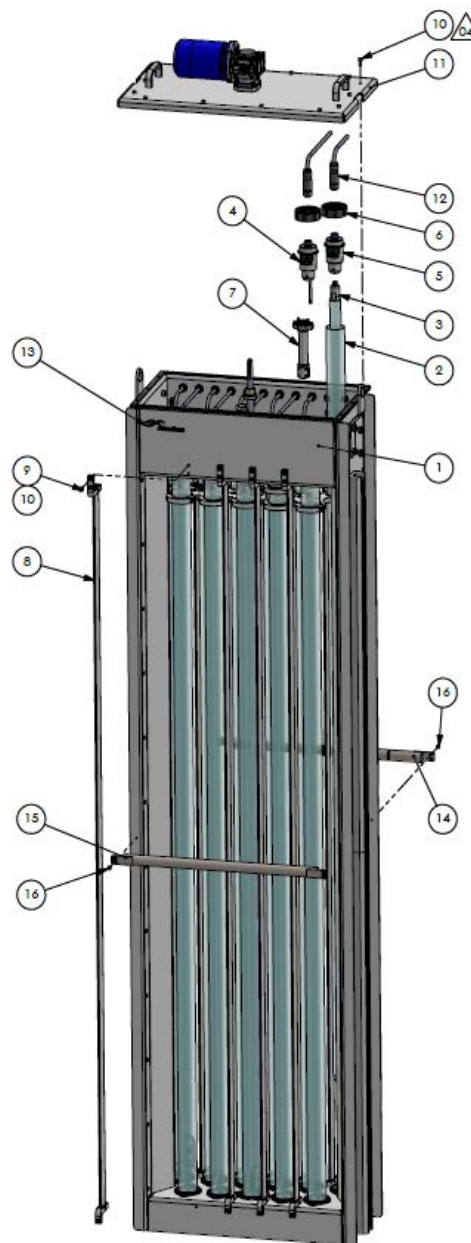


Figure 2-5: OCUVS cassette

2.1.1.1 Three Cassette Width Configurations

The **Open Channel UV System** is available in three channel-width configurations: - 750 mm, 1550 mm and 2350 mm.

750 mm channel width - (Figure 2-2 on page 8)

- (Optional) 1 cassette with 6 UV lamps
- In sequence: One or more cassettes with 11 UV lamps each
- Up to 4 cassettes in sequence, UVIS & UVTS sensors are located on cassettes A & B
- Beyond 4 cassettes, UVIS & UVTS sensors are located on cassettes A & D

A channel of 1,550 mm - (Figure 2-3 on page 9)

- In sequence: 1 or more rows of 2 cassettes of 11 lamps each, side by side (no option to use a cassette of 6 lamps)
- UVIS & UVTS sensor locations: as with 750 mm channel width

A channel of 2,350 mm - (Figure 2-4 on page 10)

- In sequence: 1 or more rows of 3 cassettes of 11 lamps each, side by side (no option to use a cassette of 6 lamps)
- UVIS & UVTS sensor locations: as with 750 mm channel width

2.1.2 Lamp Drivers Electrical Cabinet

This section describes the Electrical Cabinet that includes the lamp drivers.



Figure 2-6: LDEC cabinet – external view, showing green Lamps On and red Fault lights

The Lamp Drivers Electrical Cabinet (LDEC) Figure 2-6, which supplies power to the UV unit, is described in the following table:

Table 2-2: LDEC Physical Details

Item/Parameter	Details*
Dimensions (mm/inch): Width: Depth: Height:	612/24.1 567/22.3 1899/74.7
Max Weight (Kg/lb)	320/705
Minimum Area for Opening: Door (mm/inch): Clearance Radius: Side Clearance:	105° 589/23.2 124/4.9
Protection Standard	IP 54

*Values appearing in this table are for reference only and subject to change.

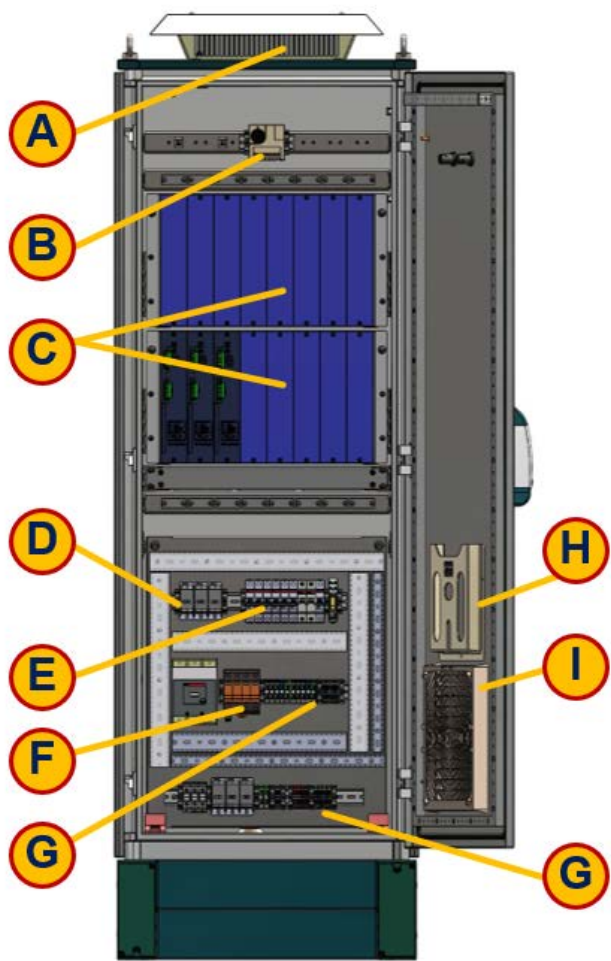


Figure 2-7: LDEC - internal view

Table 2-3: LDEC – legend

#	Description
A	Top cooling fan
B	Thermostat
C	Lamp driver units slot cover (where no driver is installed)
D	Distribution block
E	Circuit breakers
F	Surge protector
G	Terminal block
H	ICD files
I	Door fan

2.1.3 The Lamps Driver

Table 2-4: Lamp driver physical & electrical details

Item	Details
Atlantium P/N	PSW001000
Description	2x1000W LP (Low Pressure) Lamp Driver Each 19" rack contains eight 2x1000W LP Lamp Driver units to power 16 lamps in total. The LDEC contains one or two 19" racks, powering up to 32 lamps.
Dimensions of 19" 8 unit rack (H x W x D)	483 x 266 x 404 mm / 19 x 10.5 x 15.9 inch
Weight of 19" 8 unit rack	32 Kg/ 70.5 Lb
Input voltage / current	1 phase (line – neutral) 180 – 305 Vrms / 6.9 – 11.8 A max
Output power / voltage range	300 – 1000 W / 105 –150 Vms
Lamp current	2 – 9.5 A
Operating temperature	0 to 50° C / 32 to 122° F
Lamp cable length max	30 m / 98 ft

2.1.4 Main Control Cabinet

The Atlantium **Open Channel UV System** includes a network of sensors, micro-switches and control instrumentation for assuring that the Type Approval is met. All of the subsystems are connected to the Main Control Cabinet, which houses the PLC processor and the 10-inch HMI screen, see Figure 2-7 on page 13.

Refer to the electrical diagram for the Main Control Cabinet, in the ICD supplied with the system, for precise details.



Figure 2-8: Control cabinet – external view, showing control screen and indicator lights

Table 2-5: Main Control Cabinet Details

Item	Details*
Dimensions (mm/inch): Width: Depth: Height with Mounting Points: Height without Mounting Points:	 994 / 39.1 518 / 20.4 1304 / 51.3 1200 / 47.3
Weight (Kg/lb):	150 / 330.7
Minimum Area for Opening Door (mm/inch): Clearance Radius: Side Clearance:	 105° 789 / 31.1 400 / 15.75
Bottom Minimum Clearance (mm/inch):	300 / 11.8
Mounting Points (mm/inch): Between Width: Between Height: Bolt Size:	4 corner mounting brackets with 4 bolt holes each 909 / 35.79 (furthest mounting bolt on mounting brackets) 1037 / 40.83 11 / 0.43
Protection Standard	IP 54
Controller type	1586X20CP

Table 2-5: Main Control Cabinet Details

Item	Details*
Communications card	Modbus RTU model 1030X20CS
Sensors connected to 1800 card	UVIS on OCUVS cassette – Inside UV lamp sleeve, monitoring UV output UVT sensor on OCUVS cassette – monitoring UV lamp through water, to measure transmission through water DPM upper limit sensor DPM lower limit sensor Position sensor of (optional) sluice gate/weir Flow Meter sensor Water Level Meter sensor
Drivers connected to 1800 card (TBD?)	Lamp drivers DPM motor driver Driver of (optional) sluice gate/weir

*Values appearing in this table are for reference only and subject to change. Refer to Mechanical ICD

Table 2-6: Main Control Cabinet – Features

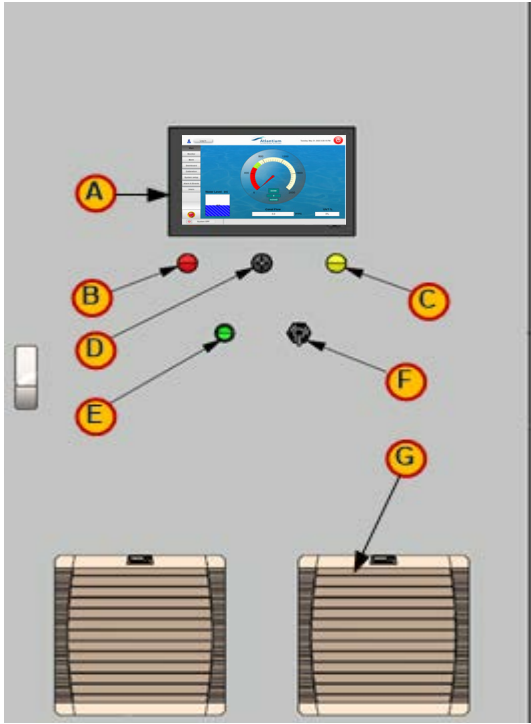
#	Item	Image
A	PLC HMI screen	 Figure 2-9: Main Control Cabinet External Features
B	RED light indication of critical alarms	
C	YELLOW light indication non critical alarms	
D	AUDIBLE buzzer for critical alarms	
E	Mute button silences the audible alarm (does not turn off the critical alarm) For a full list of critical + non critical alarms, see Table 13-4.	
F	USB port – used for downloading logs from system	
G	Door cooling fan	
H	400/440 VAC Circuit Breakers	(See Figure 2-10, below)
I	Thermostat	

Table 2-6: Main Control Cabinet – Features

#	Item	Image
J	24 VDC Circuit Breakers	
K	CPU	
L	Digital I/O cards and connectors	
M	24 VDC Power Supply	
N	Main Contactor	
O	DPM Contactors	
P	Terminal Blocks	

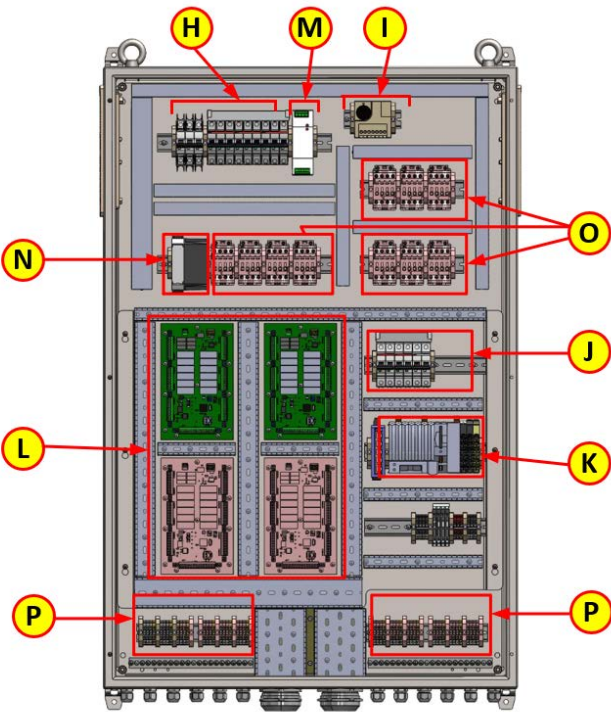


Figure 2-10: Main Control Cabinet – Internal Features

2.1.5 UV Lamps

The Low Pressure UV lamps used in the [Open Channel UV System](#) have the following characteristics:

Table 2-7: LPUV lamp characteristics

Item	Details
Atlantium PN	OPW000200
Total Length (including pins)	2211 mm / 87 in
Diameter	32 mm / 1.25 in
Electrical power	1000 W

Table 2-7: LPUV lamp characteristics

Item	Details
Nominal current	7 A
Nominal HF voltage	142 V
Start sequence	Start sequence (pre-heat) at 6A, 1100V, preheating for 12s at HF 140V
UV optical power	320W at 254 nm
Power density	2070 uW/cm ²

2.1.6 DPM system

The Deposit Prevention System (DPM) includes an electric motor (Figure 2-11) on the top of the lamp unit that rotates a screw shaft (see item (2) in Figure 1-1 on page 3), which in turn moves the DPM wiper rings (Figure 2-12 on page 18) up and down the outer sides of all the quartz sleeves.



Figure 2-11: DPM motor

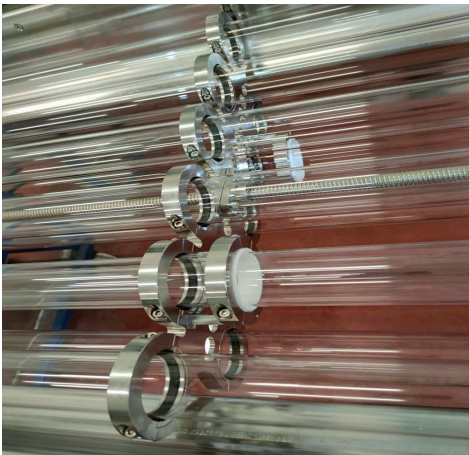


Figure 2-12: DPM Wiper Rings

2.1.7 Sluice Gate - aka Weir (optional)

An optional sluice gate (which is referred to as a “weir” in diagrams and in the soft ware HMI) may be installed to control the flow of water through the lamp cassettes of the OCUVS.

The three alternative configurations are:

- No sluice gate
- Manual, mechanical sluice gate (Figure 2-13 on page 17).
- Electro-mechanical sluice gate, controlled from X5 control cabinet, according to inlet water flow, water transparency (measured by UVS sensors)



The system may be integrated with a Sluice Gate that is provided by the customer. If The Sluice Gate is provided by the customer, it should be confirmed by Atlantium. Also, a gate provided by the customer cannot be controlled by the system.

2.1.7.1 Controlling Weir Position

Changing the height of the weir can only be performed after the water level exceeds the minimum height (**Min Weir Height**).

When the water level again falls below **Min Weir Height**, control of the weir will stop.

- The degree of opening of the weir will be carried out with the help of PID control, according to the water level.
- Opening and closing the weir will be controlled by an analog signal that is determined by the water height.
- The signal to open the weir continues until the Water Level > Top Water level, at which the weir will stop opening.

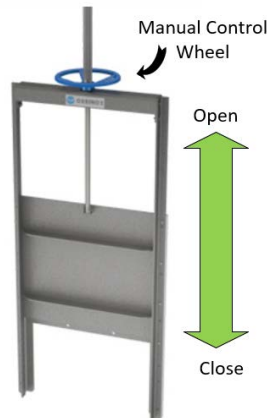


Figure 2-13: Weir with Control Wheel

2.2 Inputs/Outputs

All inputs and outputs are connected through Atlantium’s 1800 card using RTU communication.

2.2.1 Inputs

- Analog water height sensor (4-20)
- Analog Flow Meter Sensor (4-20)
- Analog UVT Sensor (4-20)
- Analog UVI sensor (4-20)
- Two digital sensors for each DPM cleaner Data received from Lamps’ drivers

- Weir's analog hight sensor (4-20)

2.2.2 Outputs

- Weir's analog control command
- Lamps On/Off control (each power supply controls two lamps)
- Lamps power control
- DPM up/down control
- Indicators of the system's status (LED indicators on the cabinet door)
- Drivers cabinet fans On/Off control

2.3 Electrical Usage

The capacities required for power source, which must be supplied to the OCUVS at the customer site, are shown in the following table:

Table 2-8: Power requirements

Electrical	Details
Main power source	400 VAC 3-phase + N + PE 50/60Hz
Current requirements per phase	determined by the number of lamps used by the system
Maximum allowed voltage tolerance	±10% from the nominal voltage

2.3.1 Connectivity

Figure 2.3.2 on page 22 shows the block diagram of for a 22-lamp system with 2 UV cassettes arranged in series, in a 750mm channel.

- The lamps are arranged in 2 cassettes/banks called A and B
- The lamps are connected to the lamp drivers in the LDEC
- Sensors, including the DPM limit switches, UV sensors, Flow meter, Level meter and weir sensors, are connected to the Control Cabinet.
- LDEC and Control Cabinet are powered from the mains supply.



Each system configuration is provided with its specific diagrams.

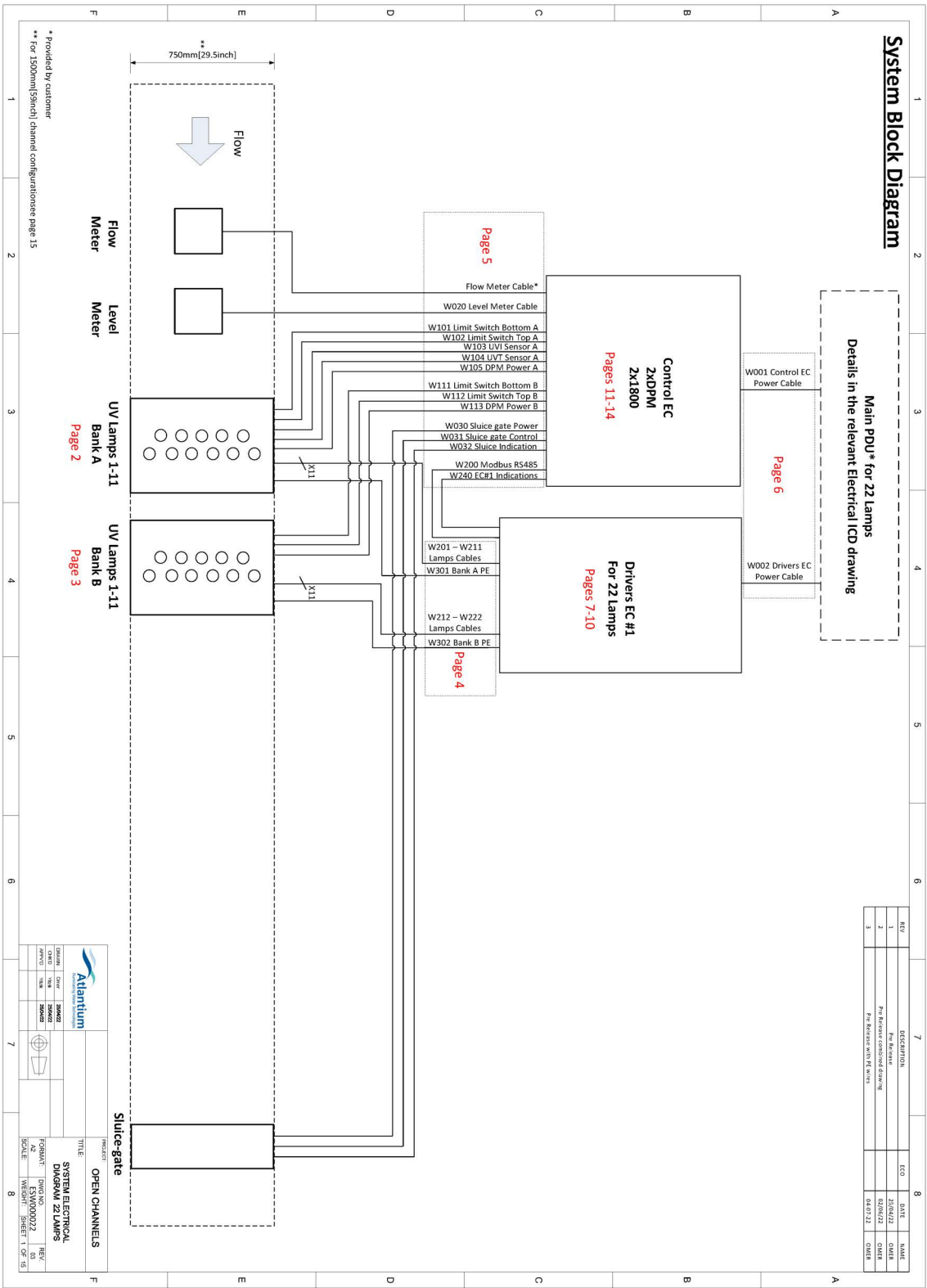


Figure 2-14: Block diagram of an example of OCUVS - for a 750mm channel with 2 UV cassettes, 11 lamps each - NB: refer to ICD supplied with your SPECIFIC delivered system

2.3.2 Power consumption

The following table shows power consumption for a 22 lamp system:

Table 2-9: OCUVS power consumption

# of lamps :			2				
Nominal lamp power (kW) :			2				
Cleaner type	Feeding type	Feeding voltage (V)	Total current per phase (A)	Main Circuit Breaker	Cabinet #1 Circuit Breaker	Controller Circuit Breakers (A)	UV Unit current per phase (A)
DPM	Direct Feeding w/o transformer	400/50Hz + Neutral	48.2	3x70A	3x60A (46.2A, 22KW)	3x10A (2A, 1kW)	48.2

2.4 Environmental Requirements

The Atlantium system LDEC and Control Cabinet cabinets are to be housed indoors in an area that is adequately cooled and with sufficient air flow. Table 2-10 indicates environmental limitations. Consult your Atlantium representative to choose the most suitable location.

It is recommended that the open channel section where the OCUVS cassettes are installed be shaded from direct sunlight and rain to protect the equipment and cabling, and also to enable lifting and servicing of the cassettes.

Table 2-10: Atlantium System Environmental Details

Environmental	Details
Maximum ambient air temperature:	45°C (113°F)
Maximum ambient air temperature in the Lamp drivers and the control cabinet environment:	45°C (113°F)
Maximum atmospheric relative humidity:	90%
Water Temperature: Maximum – high temperature: For water temperature above 70°C (160°F) consult with Atlantium service engineers Minimum – cold temperature:	30°C (86°F) No limit (no freezing)

3 Safety Overview

3.1 Before You Begin – Safety First

The Atlantium system LDEC and Control Cabinet cabinets are to be housed indoors in an area that is adequately cooled and with sufficient air flow.

It is recommended that the open channel section where the OCUVS cassettes are installed be shaded from direct sunlight and rain to protecting the equipment and cabling, and also to enable lifting and servicing of the cassettes.

Be sure to take all necessary precautions during use and maintenance.

Before you begin using this product, or any installation or service operation, please read the following safety information:

- Attention to these warnings helps prevent personal injuries and damage.
- It is your responsibility to use the product in an appropriate manner.
- You are responsible if the product is used for other than its designated purpose or in disregard of Atlantium instructions. Atlantium shall assume no responsibility in any way for such use of the product.
- The product is used for its designated purpose if it is used in accordance with its product documentation and within its performance limits.
- Using the product requires technical skills. It is therefore essential that only skilled and specialized staff or thoroughly trained personnel be allowed to use the product.
- Keep the basic safety instructions and the product documentation in a safe place and pass them on to the users.
- Applicable local or national safety regulations and rules for the prevention of accidents must be observed at all times.

The Atlantium System has been designed according to the highest safety standards, assuring the safety of operating personnel, the environment and the treated water.

3.2 Tags and Their Meaning

The following indicators are used in the product documentation to warn the reader about risks and dangers.



DANGER!

Indicates a hazardous situation which, if not avoided, will result in death or serious injury.



WARNING!

Indicates a hazardous situation which, if not avoided, could result in death or serious injury.



CAUTION!

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury.



ATTENTION!

Indicates the possibility of incorrect operation which can result in damage to the product.



Indicates a hazardous situation involving electricity which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving Electrostatic Discharge (ESD), which, if not avoided, can result in damage to the product.



Indicates a hazardous situation involving UV light which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving a hot surface which, if not avoided, can result in death or serious injury.



Indicates a hazardous situation involving working in an enclosure with moving parts, which if not avoided, can result in serious injury to the hands.



Indicates a hazardous situation involving mercury, a poisonous material which, if not avoided, can result in death or serious injury.



Indicates that components or equipment are heavy and care is to be taken to avoid lifting incorrectly. Incorrect lifting can be dangerous to the personnel lifting and may result in dropping and damaging the components or equipment.



Indicates that components or materials to be discarded are classified as hazardous waste and must be disposed of appropriately.



Indicates information related to installation, safety or use of system.

DANGER!

Improper use of controls or adjustments or performance of procedures other than those specified herein could result in significant hazards.

Therefore, personnel operating or servicing this system must be thoroughly familiar with all safety requirements and operating procedures and are to adhere to them during use of this system.



To guard against injury, observe basic safety precautions, including the following:

All persons operating the Atlantium system must be aware of proper use and the potential hazards of violating safeguards. Be certain that all personnel carefully review the safety information and the procedures specified in this manual. Only authorized individuals with appropriate training and knowledge, including local regulations, should operate, assist in the operation of or provide maintenance to the system.

All service and repair of the Atlantium system must be performed by Atlantium field service technicians or other factory-authorized personnel **ONLY**.

3.3 UV Lamp Safety



UV Exposure: The system generates ultraviolet (UV) light, which can cause serious eye damage or blindness if you stare at it directly when it is operating. Do not look directly at the lamps during system operation or when examining or servicing the system's components during operation.

Persons potentially subject to UV light exposure – technicians who are servicing the Unit – must wear appropriate eye protection whenever the system is operating.

During maintenance procedures that are conducted with Place Caution the UV lamps turned on:

- TBD appropriate safety for LP UV
- Wear white cotton gloves when handling lamps.
- Wear long sleeves rolled down to your wrist near the lamps.
- Wear appropriate eye protection, such as containing polycarbonate lens that meet EN1 ultrasonic, CAN/CSA-Z94.3-02 and/or Z87.1 standards.
- **DO NOT look directly at UV light.**



Protect Hands: Do not touch the UV lamp with bare hands.

- Use appropriate protective gloves both to protect your hands and to avoid skin oils that leave fingerprints and/or harm the UV lamp.
- Keep materials that are sensitive to heat, or which contain solvents, a safe distance away from the lamp, Unit, and electrical connections.



Electric Shock:

Before replacing a UV lamp or other components, and during any maintenance requiring lamps to be turned off, make sure the the Connection Boxes of the relevant lamps are turned OFF and place a sign on the alerting others NOT to touch the during maintenance so that no one can inadvertently turn it on while maintenance is in progress.

ATTENTION!

Prevent Damage:

Handle the UV lamp assembly by holding only the contact housings (white ceramic casing).

When handling the UV lamp, always place it on a flat surface or table, so that it is not accidentally damaged.

If the body of the UV lamp is touched accidentally, clean the fingerprints off with alcohol (at least 70%) and wipe dry with a soft, clean, lint-free cloth (usually provided with the lamp). Do not use cleaning rags or materials that can leave a residue.

Do not use a UV lamp that shows any scratches, cracks, or other damage.



Mercury Poisoning:

Lamps contain mercury, which is a hazardous substance. Always wear gloves and safety glasses when handling or replacing a UV lamp.

Inhalation of vaporized mercury compounds can be harmful to the lungs, kidneys, and nervous system. Mercury that penetrates the skin or is ingested can also be harmful.

If mercury is inhaled, penetrates the skin, or is ingested accidentally, seek emergency medical treatment immediately.

If a UV lamp breaks during handling and releases mercury, the following precautions should be observed to minimize the risk of exposure to mercury:

- Always wear gloves and safety glasses when handling or replacing a UV lamp.
- Leave the area immediately to avoid inhalation of the mercury vapor.
- Thoroughly ventilate the area for at least 30 minutes or until the mercury vapor concentration is in compliance with applicable federal and local health and safety regulations.
- After handling a broken UV lamp, carefully remove gloves, and then wash hands thoroughly with soap and water. Follow all applicable federal and local health, safety, and environmental regulations.
- Do not turn on or operate the system until all mercury contamination has been cleaned up and removed.
- Establish a Hazard Response Plan to deal with all related hazards and provide it to all personnel associated with the Atlantium system.

Refer the Chemical Safety information for mercury at: <http://www.inchem.org/documents/icsc/icsc/eics0056.htm>

ATTENTION!

Lamp Compatibility: Use original Atlantium UV lamps ONLY. Other lamps are incompatible. They void the warranty and may damage the system.



Proper Disposal of Lamps

Expired lamps are classified as hazardous waste and must be disposed of appropriately.

- A non-ruptured UV lamp that is no longer used must be disposed of according to specific local or federal environmental and hazardous waste regulations.
- Broken lamps and parts contaminated by the mercury have to be vacuum-packed and disposed of. Broken lamps, as well as contaminated packing material and other parts have to be considered as special waste, which may only be removed by authorized waste disposal companies.

3.4 Electrical Hazards and Safety Considerations

DANGER!

UNDER NO CIRCUMSTANCES

SHALL ELECTRICAL WORK BE CARRIED OUT WITHOUT VERIFYING THAT ALL MAIN CIRCUIT BREAKERS OF THE ELECTRICAL LINE FEEDING THE ATLANTIUM SYSTEM COMPONENTS ARE SET TO THE **OFF** POSITION AND THAT **NO** ELECTRICAL FEED IS LIVE.



DO NOT SET ANY MAIN CIRCUIT BREAKER OF THE ELECTRICAL LINES FEEDING THE ATLANTIUM SYSTEM COMPONENTS TO THE ON POSITION UNTIL **ALL** ELECTRICAL WIRING IS PROPERLY CONNECTED. UNCONNECTED WIRES/CABLES IN A LIVE ELECTRICAL FEED IS A HAZARDOUS SITUATION THAT CAN RESULT IN DEATH OR SERIOUS INJURY.



The Atlantium OCUVS incorporates high-voltage internal components, which can cause serious injury or fatal electrical shock if not used or serviced properly. High voltage components can retain a charge for some period of time even after the system has been turned off.



- Never attempt to remove any system covers or to dismantle any parts, except when performing the maintenance procedures detailed in *System Maintenance* on page 108.
- No portion of the Atlantium system is to be opened or removed by anyone other than a trained and authorized technician.

■ Add precautions for outdoor use EDIT TBD

- Do not spray or pour any type of fluid directly into the Unit or electrical components. Moisture causes damage to the equipment and electrical shock may result.
- Do not operate the system if the power cables or harnesses are frayed or otherwise damaged.

CAUTION!



DO NOT enter **Technician Mode** in the **Control Module** unless instructed to do so by Atlantium certified personnel or by an Atlantium System User Manual, for a specific task.

If you are in Technician Mode, **DO NOT WALK AWAY!**



NEVER LEAVE THE ATLANTIUM SYSTEM WHEN IT IS OPERATING IN TECHNICIAN MODE!!

A SYSTEM LEFT IN TECHNICIAN MODE WITHOUT SUPERVISION CAN RESULT IN PERMANENT DAMAGE!!



Utilize best practices regarding electrostatic discharge (ESD).

Avoid direct contact with the electronic circuit boards of the **Control Module**, which is sensitive to electrostatic discharge (ESD).

Wear an electrostatic discharge (ESD) Wrist Strap when performing procedures where contact with a circuit board is possible.

3.5 System Safety Features

The Atlantium Hydro-Optic Water Disinfection system includes the following safety features:

3.5.1 UV Protective Seal



While servicing the UV Unit or checking the Intensity sensor, UV blocking safety glasses should be worn by service staff or others working in the area to prevent exposing the eyes to the UV light.

3.5.2 Software Safety Controls on Control Module



CAUTION!

When the system is started, the Atlantium software performs a series of internal self-testing routines prior to starting system operation.

If the system **does not** pass these tests satisfactorily, UV lamp operation is disabled until the problem is resolved.

If problems arise during normal operation, appropriate error messages with varying levels of severity are generated on the touch-screen of the Control Module.

Control Module screens that provide setup options and compliance reports are password protected.

ATTENTION!

DO NOT leave the Control Module unattended, even for a short time, while setting change screens are accessible. An untrained person could damage the operation of the Atlantium system by making unauthorized setting changes.

3.6 Atlantium Training

As part of the Commissioning and Hand-Off process, Atlantium provides hands-on training on all aspects of the Atlantium system for your designated personnel. Should any component be replaced with upgraded features, Atlantium offers training as needed and is ready to provide additional training for new personnel or any other needs related to its systems.

4 Installing the Open Channel UV System



The installation instructions given in this manual must be followed. Failure to do so may render the warranty void.

Read through the following installation section **before** you begin the installation process.

4.1 Installation overview:

Hardware setup - to be implemented by your facility personnel:

Phase 1 Unpacking the Atlantium OCUVS and checking the delivered items (See below page 32).

Phase 2 Installing the frames for the OCUVS cassettes into the concrete channels, including creation of designed slopes in concrete edges adjacent to frame to optimize water flow through the cassettes.

Phase 3 Inserting the OCUVS cassettes into the frames, within the concrete channels, using the cranes.

Phase 4 Installing the **Control Cabinet** and LDEC, etc. (p 37)

Phase 5 Cabling the **Control Cabinet**, LDEC, etc. (p 41)

Phase 6 Installing the Drivers Cabinet

Phase 7 Commissioning - to be implemented by Atlantium personnel:

Phase 8 Starting up the system

Phase 9 Configuring the **Control Cabinet**

Phase 10 First time System Activation

The Installation Flow diagram below maps out the installation process.:

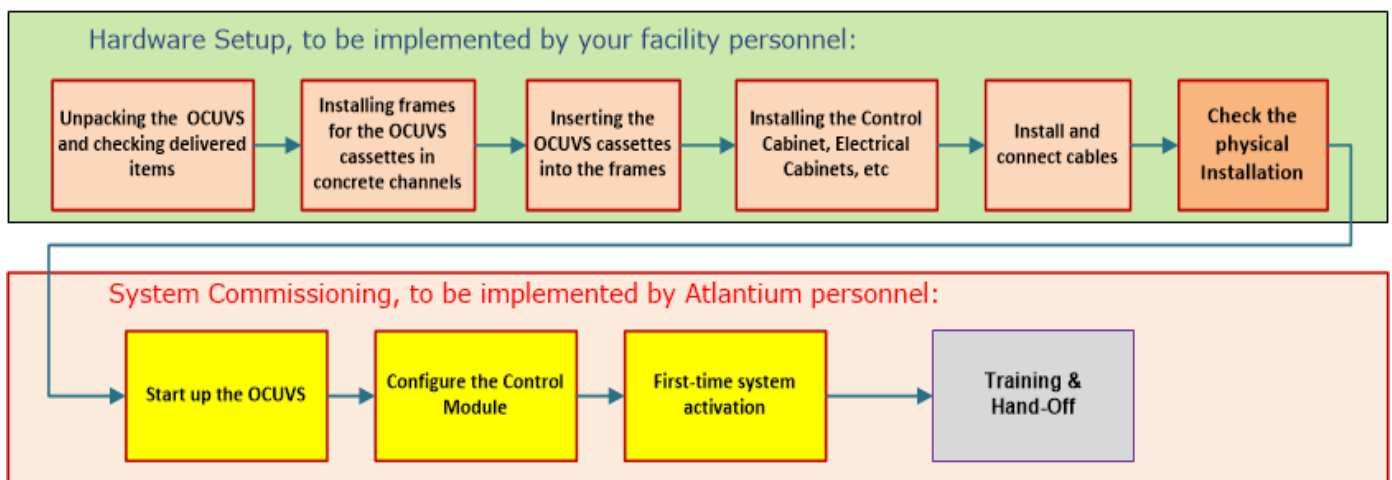


Figure 4-1: Installation Flow Diagram

4.2 Supply Conditions

4.2.1 Equipment Responsibility

Upon receipt of the equipment the system owner or installation yard automatically becomes fully responsible for the equipment, unless otherwise stated in the contract. This responsibility covers the storage period before installation, the actual installation, commissioning, and the period between the completion of the commissioning and the acceptance of the equipment by the end user (normally the owner of the vessel into which the equipment is to be installed).

4.3 Storage Guidelines

In some cases, it is inevitable that the system must be stored for a certain period of time. When this is necessary, make sure that system components' transport cases are stored somewhere **protected from direct sunlight, dry and clean**. High humidity, salty air etc. may noticeably reduce the life time of equipment when not connected and operational and not maintained. We recommend not to exceed a storage time of four months.

Caution!



There is a risk that system components may be damaged due to improper storage.

The transport case must not be stored outdoors.

Protect the system from moisture and contamination during unpacking and storage.

Do not store the system in freezing conditions.

4.4 Unpacking And Checking Package Contents

The Atlantium system is packaged and shipped in specially packed crates that contain:

- The **OCUVS** cassettes
- The **OCUVS** cassette frames
- The LDECs
- The Control Module
- Cabling
- Documentation
- Peripheral accessories as per your order and listed on your delivery documents are crated according to their size and weight. Small accessories may be crated together.

Each of the crates is connected to a forklift-friendly wooden pallet. The Atlantium crates must be lifted off the transport vehicle by a qualified operator, using a forklift or crane

➔ To unload and check the system:

- Step 1.** Using the forklift or crane (not supplied by Atlantium), lift each of the crates off the transport vehicle and lower them to a level ground surface.
- Step 2.** Walk around and inspect each crate for damage. Note if there is any separation in the joints or any other indication of any damage in transit. If there is any indication of damage, take a photograph of it and use extra caution when proceeding to the next step.
- Step 3.** Un-clamp and remove the panels of the crates.
- Step 4.** As you remove the wrapping and shipping protections, verify that the contents of the crates are correct according to the delivery documents. The **OCUVS** Unit contains a plate with the model number and other product information. Verify that the correct product version is supplied.



- Immediately report to Atlantium and to the insurance carrier any damage to the crates or to the Atlantium system found before opening the crates or during unpacking, installing or testing of the system.
- If you ordered more than one Atlantium system, check the serial numbers against the delivery documents to verify which components comprise each individual system.

4.5 Before You Start

Before you start the installation, take the following actions:

- Inform the supervisor / coordinator that the work is about to be carried out.
- Collect the required Installation Design Plan, drawings, and other documentation and read the applicable procedures before commencing work.

4.6 Installing the OCUVS frames



At all times, the bank cassettes must be lifted using a specially prepared lifting bar, as explained in **TBD**

Each OCUVS cassette is to be lowered from above into a slide-in metal frame, to be installed in channels with concrete walls (Figure 4-2, on page 34).

The metal frames, (shown in green outline in Figure 4-2, on page 34) which are supplied to measure by Atlantium, must be embedded at the designated positions inside the concrete channels.

Two widths of metal frame shall be used, depending on requirements:

- Single-unit frame 750 mm (29.5 in) wide and 2300 mm (90.5 in) deep, for single 11-lamp cassette width channel
- Double-unit frame 1500 mm (59 in) wide and 2300 mm (90.5 in) deep, for double 22-lamp cassette width channel



WARNING!



The OCVUS frames and UV units are very heavy. Use cranes to lift them from packaging and to slide into place in frames.

The UV lamps and quartz tubes are fragile. The cassettes must be handled with great care to avoid bangs/jolts while unpacking or moving or lowering into place.



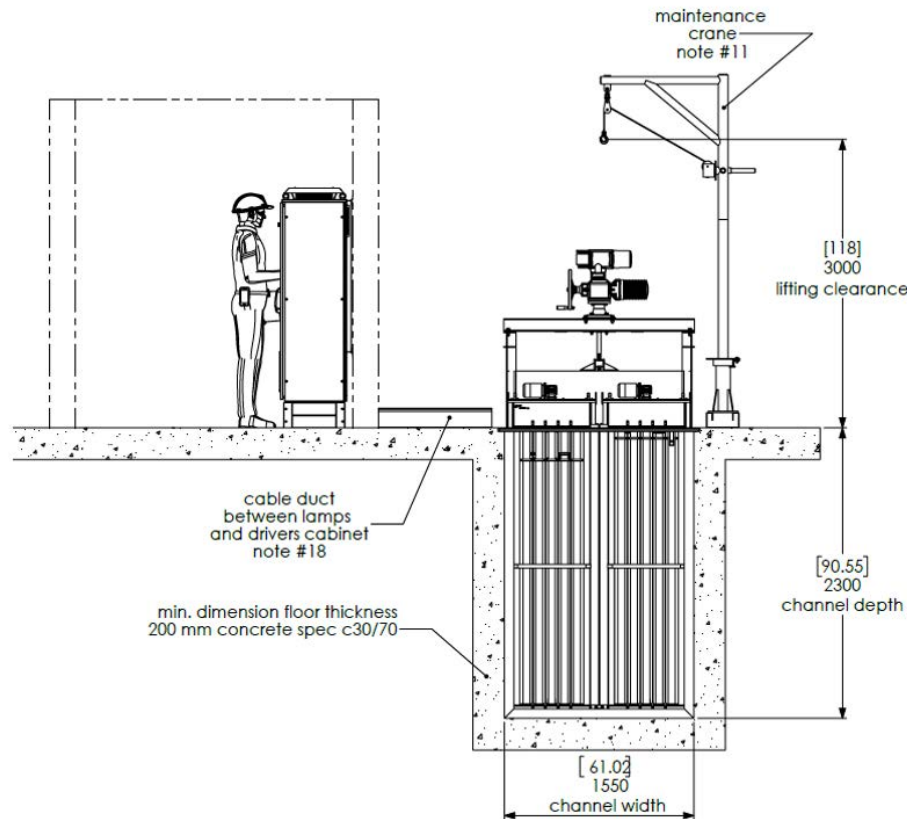


Figure 4-2: Set-up schema - Example only, refer to ICD supplied with your SPECIFIC delivered system

4.6.1 Frames Installation Guidelines

It is the responsibility of the Customer Site Manager to supply proper cranes to lower and raise the equipment, and to ensure that the frames for the OCUVS cassettes are installed and embedded securely into the concrete channels, in accordance with the agreed site plans and mechanical ICD files supplied with the system.

- The frames must be installed vertically.
- The frames must be installed in the correct upstream/downstream orientation, in order to optimize flow performance. Check orientation on mechanical ICD files.
- While casting the concrete, be sure to create designed slopes in concrete edges adjacent to frame to optimize water flow through the cassettes.
- Care must be taken to avoid any distortion/bending of the frames during embedding into the concrete channels, otherwise the OCUVS cassettes may not slide in/out properly!
- All points of contact between the frames and the concrete walls must be properly sealed, to avoid any flow of untreated water around the frames.
- The concrete seals around the installed frames must be allowed to dry and harden thoroughly before the open channels are filled with water.
- All necessary building practices and measures must be taken to ensure proper installation of the frames.

4.7 Placing the OCUVS Unit

Make sure to use the Mechanical ICD provided with the system during this procedure. To anticipate the correct height for transport, trace the path to be taken to transport the Unit to its designated location and clear it of all obstacles.



WARNING!

There is a risk of accidents from falling loads.

During installation of the system parts, accidents may occur due to sizes and weights.

- Use correctly proportioned and intact lifting accessories.
- Attach the separate lifting accessories properly to the components.
- Do not stand and do not work under raised loads.
- Wear the appropriate safety equipment and clothing (helmet, protective gloves, safety shoes, etc.)

4.7.1 Required Equipment

The following tools and equipment are required:

- Lift or crane
- Lifting strap/shackles with latch hooks



The **OCUVS** unit is heavy – refer to TBD. Two or more people are required to get the **OCUVS** Unit to the designated installation location – a qualified lift or crane operator and at least one other person to guide the Unit as it is transported.

➔ To place the OCUVS Unit in embedded frame:

- Note that the **OCUVS** Unit is to be installed vertically (see **Figure 4-3**).
 - The Unit has 2 screw-in lifting eye support points **(A)** & **(B)**.
- Step 1.** Ensure that lifting eyes **(A&B)** are screwed into the support points on upper part of unit.
- Step 2.** To lift the Unit from its packing crate, use a horizontal bar **(3)** connected to the lifting beam **(5)** of the lift or crane to attach vertical lifting straps **(1 & 2)** with latch hooks through the lifting eyes **(A&B)** on either side of the upper part of the unit.
- Step 3.** DO NOT USE triangular strap alignment as shown on right.
- Step 4.** Raise the unit so that it remains steady and vertical.

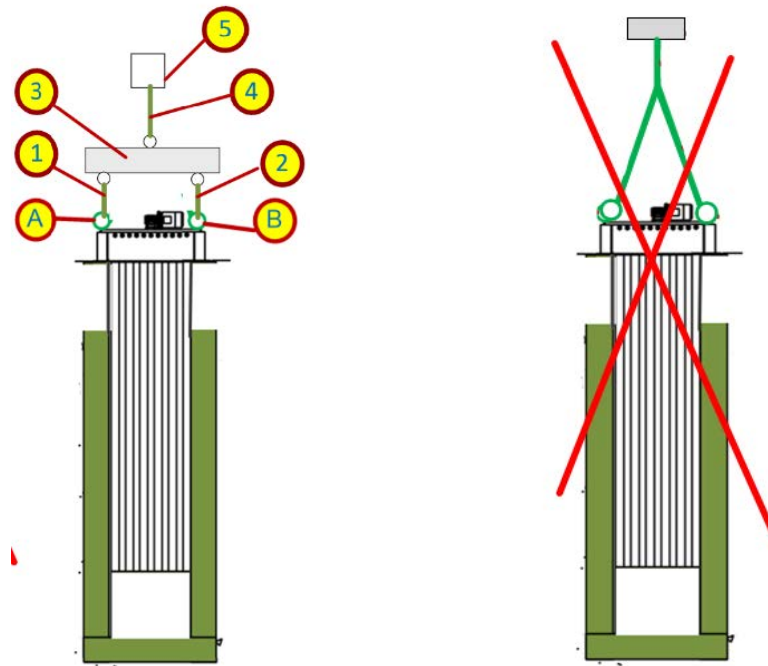


Figure 4-3: Configuration for lifting

- Step 5.** Ensure that all straps are connected securely to the Unit, to each other and to the lift/crane.
- Step 6.** Using the lift/crane, lift the Unit by the straps, out of the crate and position it for safe transport (near floor level is preferred).
- Step 7.** Using the lift/crane and the designated person(s) to guide, carry the Unit to its intended location.
- Step 8.** At the designated location, check the installation plan, including the distances between the bolt holes, etc.
- Step 9.** Lower the unit slowly into the embedded frame (Refer to the provided **Open Channel UV System Interface Control Drawing**.)
- Step 10.** The same principle is to be used to remove the unit from its frame for cleaning/maintenance.

4.8 Power connections



- The wiring and connection of all system components are to be performed on site.
- It is very important that cables connected to the cassettes (lamp power and sensor cables) are planned and arranged so as to allow safe disconnection and reconnection for the raising of the cassettes, changing of lamps and other servicing /maintenance procedures.
- It is vitally important to ensure that no cable connector is able to fall into the water at any time, during operation or servicing!
- Circuit breakers outside the Control Cabinet and LDECs are per the electrical ICD, and are not provided by Atlantium.

The wiring and connection of all system components are to be performed on site.

- Ensure that all power is switched off to the system.

- Label the ON / OFF switches, circuit breakers and fuses with warning notes clearly stating that work is being carried out on the system.



WARNING!

Do not attempt to run the system before the checks listed in the Atlantium check and verification procedure have been completed.

4.8.1 Setting the Main Circuit Breakers to OFF Position

Before beginning work to connect the wires/cables of the Atlantium system components, turn the main circuit breakers of the electrical lines feeding the Atlantium system components to the **OFF** position. **Verify that no electrical feed to the Atlantium system components is live.**



WARNING!!!

Under no circumstances shall electrical work be carried out without verifying that **ALL** main circuit breakers of the electrical lines feeding the Atlantium system components are set to the **OFF** position and that **NO** electrical feed is live.



DO NOT set any main circuit breaker of the electrical lines feeding the atlantium system components to the on position until **ALL** electrical wiring is properly connected. Unconnected wires/cables in a live electrical feed create a hazardous situation that can result in death or serious injury.

4.9 Installing the Lamp Driver Electrical Cabinet(s)

The location of the LDECs must be designated according to the length of the **Cables** of the **OCUVS** Unit and the designed cable paths. The standard length of **Cables** that connect the **OCUVS** Unit to its LDEC is **20 meter (65.6 ft)**.

The LDEC can be mounted against a wall, or on the floor.

4.9.1 Required Equipment

The following tools and equipment are required:

- Lift or crane
- Lifting strap



The LDEC is heavy - refer to Table **2-2, LDEC Physical Details**, in **Section 2**, page **13**. Two people are required to get the LDEC to the designated installation location - a qualified lift or crane operator and at least one other person to guide the LDEC as it is transported.



Do not position the LDEC directly on the floor. Allow sufficient room to avoid contact with the floor, which may get wet.

The LDEC includes fans and filters on the top and bottom of the door. There must be a clearance of at least 80cm (31.5 inches) in front of door for access and for proper ventilation. To review the complete clearance requirements of the LDEC, see the mechanical ICD.

➔ To position the LDEC:

Step 1. Uncrate the LDEC package.



The LDEC of the Atlantium system is heavy! – It should be lifted and carried by mechanical means and by more than one person.

- Step 2.** Thread lifting straps through the lifting eye bolts on each upper corner of the LDEC. Attach the lifting straps together and to the lift or crane.
- Step 3.** With the lift or crane, move the LDEC to its location designated according to installation plan.
- Step 4.** For a floor-mount configuration, stand the LDEC on its built-in legs.
- Step 5.** Lower the LDEC to its designated location.
- Step 6.** As appropriate, bolt the LDEC to the wall or anchoring point.

4.10 Installing the Control Cabinet

TBD

4.10.1 Mounting and Connecting the Control Module



Information contained in this User Manual is for illustration purposes only. For detailed wiring diagrams of your particular system, ALWAYS refer to the supplied System Electrical Diagrams and Electrical ICD drawings.

The **Control Module** can be mounted on a wall or suitable anchoring point. The **Control Module** comes with an electric power cable 1.6M/5.25ft. long (1.5m/4.92ft. from the cable gland) and a Data cable. It can be placed up to 50M/ 164.04 ft. from the Atlantium Unit, in a place that it is easy to reach the touch screen and read the monitoring information. For Data Cable length options, please contact Atlantium.



ATTENTION!

- When mounting the **Control Module**, take into consideration the length of the cables.
- **Do not Stretch Electrical Cables – Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.**

➔ To mount the Control Module on skids:

Step 1. At the location designated for the **Control Module** set up the skids with spacing that matches the spacing measurements for mounting orientation on the ICD drawing accompanying the documentation.

- Step 2.** The mounting hole (1) are 8.5mm (0.33") in diameter. Using the appropriate bolts and nuts, with a screwdriver, screw in the four bolts and add the nuts leaving around 5.6mm (0.20-0.24") sticking out.
- Step 3.** Uncrate the **Control Module** package and carefully lift it out.
- Step 4.** Open the door of the **Control Module** and remove the plastic covers on the four holes.
- Step 5.** Position the **Control Module** so that its four holes fit over the corresponding holes on the skids.
- Step 6.** Using appropriate bolts and nuts and with a screwdriver, screw in the four bolts adding the nuts in a diagonal pattern (one corner, then the opposite corner).

The **Control Module** contains a jumper marked 11 in the **Figure** below.

4.10.2 Control Module Connections

- Communication connectivity can be via Digital/Analog I/O's hard wired Modbus RS485 cable connections (see below) .
- You can connect signal wires for the following peripheral equipment to the Control Module, shown in block diagram form **Figure 4-4**.

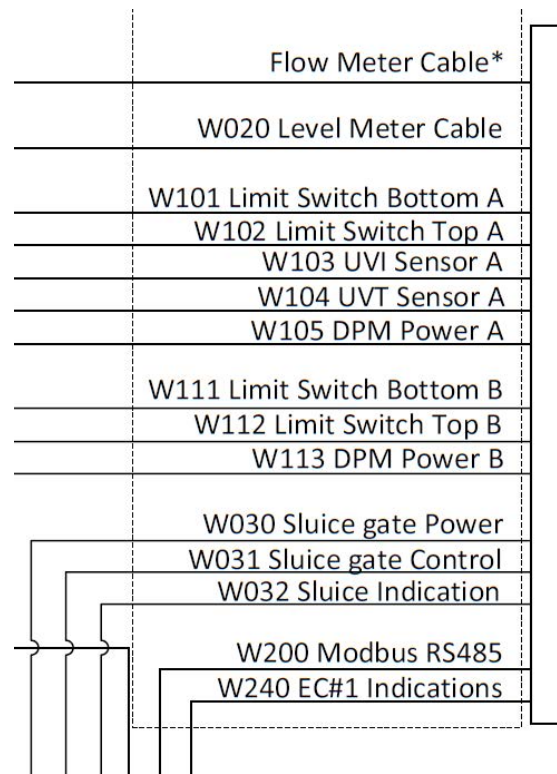


Figure 4-4: :Connections to Control Cabinet – Example only, refer to ICD supplied with your SPECIFIC delivered system

4.11 Modbus Communication Connections

Modbus is an application layer messaging protocol for client/server communication between devices connected on different types of buses or networks. Modbus is often used to connect a supervisory computer with a remote terminal unit (RTU) in supervisory control and data acquisition systems.

The Atlantium system can be controlled directly from your facility control system via Modbus – RTU on RS485 (network) serial connection.

To accomplish this:

- The Modbus Settings must be configured in the **Control Module**. See *Configuring More Settings* on page 80.
- The Facility control system communication properties must be configured to match that of the Atlantium system. See *Setting Up the Facility System Communication Properties* on page 40.
- For the terminal block connectors for the Modbus in the **Control Module**, see Figure 4-4 , on page 33 above.
- Signaling communication incoming to the facility control system is according to the Modbus Registers. .



To ensure communications are operational, be sure to connect the cables **TX to RX** and **RZ to TX**.

4.11.1

Setting Up the Facility System Communication Properties

To properly configure communications between your Atlantium system and control network, set up the communication properties and **COM1** port settings at the facility control system as follows: TBD

Node Address:	TBD
Available Commands	TBD Function codes
Function Codes: *	TBD 0x01, 0x02, 0x03, 0x04, 0x05, 0x06, 0xF, 0EC-10
Support IEEE Floating Point: **	TBD NO (register/100)
Baud Rate:	4800, 9600, 19200, 38400, 57600, 115200 ***
2 Wire	
Parity:	None TBD
# of Stop Bits:	1
# of Data Bits	8

(*) You can read all the registers in the system at once with **0x03** (includes coils, discreet coils, holding registers and input registers).

(**) Divide the desired parameter by 100 to calculate the real flow point.

(***) Must be the same in the Control Module setting



Make sure the circuit breakers before the mains are set to the **Off** position **BEFORE** connecting the Atlantium system to the mains.



Avoid tension on the Electrical Cables: **Do not Stretch Them**. They are to be installed within conduits and must not be stretched.

4.11.2 Connecting the Control Module's Power Cable

The **Control Module** comes with a power cable of 1.6m/5.25ft. length. For the electrical specifications, See ***The Control Module*** on page 22. Prepare the appropriate plug for the electrical standards of your region. Considering the distance between the **Control Module** and your electrical box, you may want to replace the electrical cable with a longer one.

4.11.3 Connecting the Cables to the LDEC

The connections to the LDEC are shown in block diagram form in **Figure 4-5**.



Make sure the circuit breakers before the mains are set to the **Off** position. **BEFORE** connecting the Atlantium system to the mains.



- Do not Stretch Electrical Cables - Avoid tension on the electrical cables. They are to be installed within conduits and must not be stretched.
-



WARNING!

- The electrical connections must be performed by a qualified electrician **ONLY**.
 - **Do not** plug in the Unit's Power cable to an outlet until after the hardware installation is complete and the system is ready to be initialized.
-

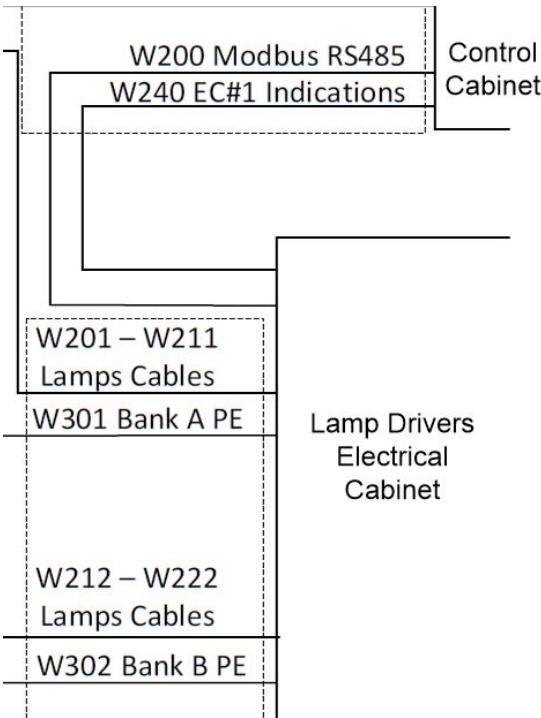


Figure 4-5: Connections to LDEC - Example only, refer to ICD supplied with your SPECIFIC delivered system

4.12 Installation Check List

When you are ready to schedule the Atlantium technician to commission the Atlantium system,

Step 1. Complete the checklist below – Table 4-1.

Step 2. Make digital photos of every angle of the installed system.

Step 3. Send the photos and the completed checklist to your Atlantium representative.

Table 4-1: Atlantium System Installation Check List

Item	Done
■ The embedded frames are installed: ☐	
■ The OCUVS Units, LDECs and other components are located in their final position.	☐
■ All of the System components are securely installed and have sufficient service clearance on all sides (according to the ICDs).	☐
■ All power preparations are complete and Power connects are wired as instructed in this guide.	☐
■ Correct voltage was measured where the Power connects to the LDEC Voltage rate _____ Number of phases _____	☐
■ A flow meter or flow switch is installed and the electrical and data connection are complete. ☐ If flow meter: 4 - 20 mA signal and Com wires are connected as instructed in this guide. ☐ If flow switch: 2 wires from the flow switch are connected as instructed in this guide. ☐ A 4 - 20 mA signal from the PLC is installed.	☐
■ The Control Module connections are completed. All supporting features (e.g., General Alarm, External On/Off, flow meter or flow switch, auto valves open/close, etc.) and wiring were installed according to the Installation plan based on the instructions in this guide.	☐
■ All LDEC connections are completed.	☐
■ You are scheduling an electrician to be on site and available during the System commissioning (a must or the installation cannot be completed!).	☐
■ To be sent to Atlantium with this checklist, attach photos of the following: ☐ OCUVS Unit ☐ LDEC ☐ Control Module and other components ☐ Accessories (Flow meter or flow switch)	
Customer Approval Name of Customer's authorized representative: _____ Position: _____ Signature of Customer's authorized representative: _____ Date: _____	

5 Open Channel UV System SETUP

5.1 System setup and calibration

This section details how to perform **Open Channel UV System** initial setup and calibration .



The setup and calibration procedures for the **Open Channel UV System** are to be performed by /under the supervision of Atlantium technicians.

Do not attempt to run the system before the checks listed in the Atlantium check and verification procedure have been completed.

5.2 Starting up

For system setup, switch on the control cabinet of the OCUVS and log in as Admin – as described in the next chapter under the Section **Starting Up** on page 53.

5.3 System Setup

On the screen of the control cabinet, press System Setup tab on left tab column to access Parameter settings, see Figure 5-1. The following table lists default initial parameter settings to be performed by authorized personnel.

System Setup Screen Details:

- Channel Configuration:**
 - Number of Banks in Channel: 1
 - Number of Lamps Per Bank: 6
 - Number of Columns: 1
 - Number of Rows: 1
- Flow Parameters:**
 - Flow Sensor Type: Flow Meter
 - Minimum Validated Flow: 20 m³/h
 - Maximum Validated Flow: 300 m³/h
 - Flow Units: m³/h
- Water Level Parameters:**

Parameter	Value	Action	Time
Maximum	1.8 m	Alarms	0.1 min
Target Height	0.0 m		
Minimum	0.9 m	Alarms	1.1 min
Critical Minimum	0.5 m	Alarms	1.0 min

Water Level Units: Meters
- Process Parameters:**
 - Process Mode: Power Mode
 - Power Target: 100 %
- DPM Parameters:**
 - Cleaning Pass Timeout: 301 s
 - Time Between Cycles: 0.5 min
 - Time Until Maintenance: 1.00 y
- Time Parameters:**
 - Communication Timeout: 30 s
 - Alarm Delay: 30 s
- UVT Parameters:**
 - Minimum Validated UVT: 20 %
 - Maximum Validated UVT: 75 %
- Dose Parameters:**
 - Minimum Validated Dose: 11 mJ/cm²
 - Maximum Validated Dose: 9,999 mJ/cm²
- Driver Cabinet Parameters:**
 - Fan Stop Delay: 5 s
 - Dose Factor: 1.00
 - Calibration Time: 2 min
 - LampType: 48
- Weir Parameters:**
 - Weir Type: Manual
 - Open Target: 60 %

Status Bar:

- Lamps On - On Spec
- Active Alarms: 1
- All Alarms: 1

Figure 5-1: System setup screen

This screen gives access for setting up the system.

5.3.1 Parameter Settings

Table 5-1: System Setup

Parameter	Description	Valid Range	Default Value
Channel Configuration			
Number of Banks in Channel	Number of Banks in Channel (according to configuration)		
Number of Lamps Per Bank	Number of Lamps Per Bank	11/6	11/6
Number of Columns	Number of Columns	1-3	1
Number of Rows	Number of Rows	1 - 10	1
Process Parameter			
Process Mode	Selects between Dose Mode and Power Mode		
Power Target (Power Mode)	Sets the required power of the lamps	50-100[%]	100[%]
Target Dose (Dose Mode)	Sets the required dose	10-9999 [mJ/cm2]	
Dose Target Delta (Dose Mode)	Approved deviation from Target Dose at Dose mode	50-100[%]	
Time Parameters			
Communication Timeout	Time out for defining if there is issue regarding communication with Driver and 1800 cards	0-120 [sec]	30 [sec]
Alarm Delay	Time pause until alarm is activated	0-120 [sec]	30 [sec]
Dose Parameters			
Minimum Validated Dose	In case the Actual Dose < Minimum Dose: System will be defined as "OFF SPEC"	10-9999 [mJ/cm2]	10 [mJ/cm2]
Maximum Validated Dose	Maximum approved value	500 [mJ/cm2]	
Flow Parameters			
Flow Sensor Type	Flow Meter / Flow Switch		
Minimum Validated Flow	Assumption	0-8000 [m3/h]	100 [m3/h]
Maximum Validated Flow	Assumption	0-8000 [m3/h]	8000 [m3/h]
Flow Unites	Flow Unites	Metric: a.m3/h b. MLD (million liter per day) US: MGD(Million gallon per day)	
DPM Parameters			

Table 5-1: System Setup

Parameter	Description	Valid Range	Default Value
Cleaning Pass Timeout	Timeout delay for DPM movement Up-Down Down-up	0-120 sec	60 sec
Time Between Cycles	Delay timer between Start cleaning cycle of DPM	1-300 [Minutes]	
Time Until Maintenance	Time based maintenance. At the scheduled maintenance time, alarm will invoke.	1 Year	1 Year
UVT Parameters			
Minimum Validated UVT	Assumption	0-100%[%-cm]	20%[%-1cm]
Maximum Validated UVT		0-100%[%-cm]	98%[%-1cm]
Driver Cabinet Parameters			
Fan Stop Delay	Sets the delay time for the fans to turn off after ALL Drivers at the cabinet are OFF.		
Dose Factor	Dose = Actual Dose*Dose Factor	0.1-100	
Lamp Type	Lamp Code number according to Lamp type	OPW000200 - 48 OPW000205 - 77	
Calibration Time	Time the lamps should be activated before the user preforms Bank Sensor calibration	1-30min	2min
Water Level Parameters			
Maximum	Maximum water level	0.7-2.3[m]	
Target Height	Water level target when Automatic Wier mode is set. Relevant only in auto mode	0.7 - 2.3[m]	0
Minimum	Minimum water level	0.7-2.3[m]	1.8 [m]
Critical Minimum	Minimum water level that triggers an action (lamp shuts off)		
Parameter	Description	Valid Range	Default Value
Water Level Unit	Water Level Unit	Metric/US Unit	
Weir Parameters			

Table 5-1: System Setup

Parameter	Description	Valid Range	Default Value
Weir Type	Control type of the Wier 1. Automatic - Via Controller 2. Manual - Fixed Opening defined at SW 3. None - No dynamic weir		
Open Target	Opening Value of the Weir - Fixed Value	0-100[%]	
Weir Parameters (Automatic)			
Max Opening	Maximum Weir gate opening	100	0[%]
Min Opening	Minimum Weir gate opening [%]	0	0[%]
Weir Delta	Approved tolerance for opening Weir [%]	0-20 %	10%
PID Proportional	Automate mode control constant variable		
PID Integral	Automate mode control constant variable		
PID Derivative	Automate mode control constant variable		

Table 5-2: Banks and Lamps

Parameter	Description	Valid Range	Default Value
Select Bank to Monitor	Selects the bank to configure		
Lamp and Bank Parameters			
Bank in Use	Checkbox is selected when Bank is in use		
UVIS/UVTS in use	Checkbox is selected when UVIS/UVTS is in use		
DPM Enabled	Checkbox is selected when DPM is in use		
Lamp in Use	Checkbox is selected when lamp is in use		
Set All in Use	Checkbox is selected when all lamps are in use		
Save Parameters	Save the configured parameters		

5.4 Calibration Screen

With the Calibration screen you can calibrate the Flow Meter and the Water level sensors.

Log Out
Current User: Admin

Atlantium
Illuminating Water Technologies

Thursday, December 28, 2023 2:39 PM

Main
Alarm and Events
Monitor
Calibration
System Setup
Maintenance

System Sensors Bank Sensors

Flow Sensor Calibration

Point	Flow	Sensor Current
First	0 m ³ /h	4.00 mA
Second	300 m ³ /h	20.00 mA

Set Calibration

Flow Sensor Live Value

Flow	Sensor Current
48 m ³ /h	6.57 mA

Weir Height Sensor Calibration

Point	Opening	Sensor Current
First	0 %	4.00 mA
Second	100 %	20.00 mA

Set Calibration

Weir Height Sensor Live Value

Opening	Sensor Current
1 %	4.00 mA

Water Height Sensor Calibration

Point	Height	Sensor Current
First	2.0 m	4.00 mA
Second	0.5 m	20.00 mA

Set Calibration

Water Height Sensor Live Value

Height	Sensor Current
1.4 m	10.38 mA

English

System Off

Active Alarms 36
All Alarms 267



Calibration of sensors is to be performed after system installation by installation team. Each sensor type has its particular calibration settings.

5.4.1 Flow Meter Calibration

- If a Flow meter is installed, it has an output of 4–20 mA. 4 mA is defined as 0 flow, 20 mA = max flow
- If a Flow switch is installed the estimated flow is set to be displayed when water flow is detected.
- The calibration is carried out by an Atlantium authorized technician who will test the system in specific states, for example:

➔ To Calibrate the Flow Meter

- Step 1.** Open the weir (Sluice gate) to achieve a specific desired flow rate.
- Step 2.** Measure the actual flow rate obtained and the sensor reading in mA.
- Step 3.** Record these 2 values and repeat for several different flowrates.
- Step 4.** The system will set up a function of flow rate against sensor reading, to accurately measure flowrate for the control of the system.

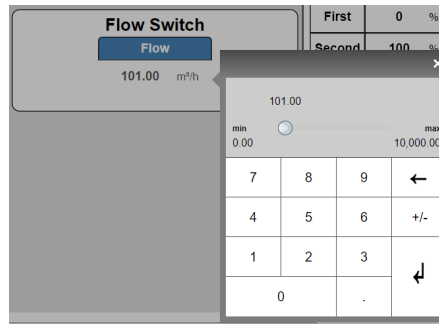


Figure 5-2: Flow Switch Setting

5.4.2 Water Level Sensor Calibration

➔ To calibrate the Water Level Sensor:

- Step 1.** The measurement from this sensor is calibrated by measuring actual water level and noting the corresponding sensor output current in mA.
- Step 2.** Repeat Step 1 above for a different water level.
- Step 3.** Enter the calibration readings. The system will create a function for reliable water level readings to be received at controller.

5.4.3 Wier Hight Sensor Calibration

➔ To calibrate the Weir Hight Sensor:

If a Wire Hight sensor is installed, then Weir Height Sensor Calibration section appears on screen.

- Step 1.** Open the Wier to a specific height .
- Step 2.** Measure the actual flow rate obtained and the sensor reading in mA.
- Step 3.** Record these 2 values and repeat for several different flow rates.
- Step 4.** The system will set up a function of flow rate against sensor reading, to accurately measure flow rate for the control of the system.

Weir Parameters	
Weir Type	Manual
Max Opening	Automatic
Min Opening	None
Weir Delta	20 %
PID - Proportional	0.00
PID - Integral	0.00
PID - Derivative	0.00

Figure 5-3: Weir Automatic Control Settings

5.4.4 UVI Calibration

➔ To Calibrate the UVI Sensor

- Step 1.** Click on Start, the lamp in the specific Cassette/bank with a UVI connected will be turned on at 100%

- Step 2.** After verifying that the lamp is 100% lit and stable, press the Calibrate button (or alternatively the Stop button to stop the process).
- Step 3.** The system will record the output current from the UVI sensor and the lamp driver
- Step 4.** The system will lower the power to 75%, 50%, 25% and after half a minute between each transition, it will collect the current value from the sensor and the driver
- Step 5.** The system will use these points to build a function of the form $y(x)=a \cdot x+b$, to convert the current value of the driver to the current value of the sensor



The function will be used for lamps that do not have a UVI sensor.

6 Operating the Open Channel UV System

This section describes how to operate the **Open Channel UV System** ;



WARNING!

Do not attempt to run the system before the checks listed in the Atlantium check and verification procedure have been completed.

6.1 Starting Up

➔ Startup and Login

- Step 1.** Switch on the main CBs in the Control cabinet (H in Figure 2-10 on page 17) and in LDEC (E in Figure 2-7 on page 13).
- Step 2.** On the Main screen of the Control cabinet HMI, tap the Login button on top LH corner (Figure 6-2 on page 55).
- Step 3.** Login by typing the appropriate user name and password on the pop-up keyboard

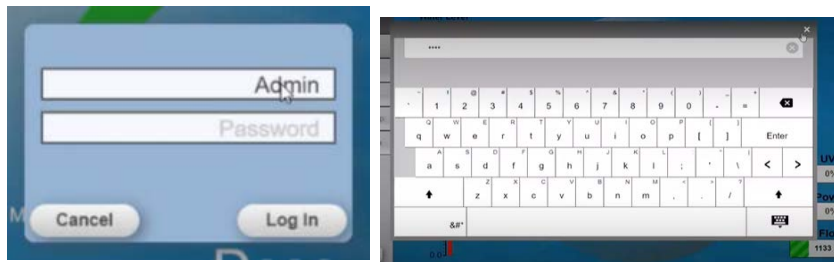


Figure 6-1: Login window and pop-up keyboard

- Step 4.** Select language (current options: English/Portuguese)
- Step 5.** Press  to light lamps.

6.1.1 User Login

There are 3 levels of user for the **RZ163**: Admin, Manager and Operator.

The following Table 6-1 shows authorizations:

+ indicates View and Edit

V indicates View only

- indicates no View or Edit

User names and password will be issued at system setup.

Table 6-1: User authorizations

Authorization	Admin	Super User	Manager	Operator
Turn On/Off	+	+	+	+
Alarms &Event log	+	+	+	+(No Clear / Export)
Monitor	+	+	+	+
Calibration	+	+	+	-
System setup	+	+	+	
Maintenance	+	-	-	-
Setup parameters				
Process	-		+	-
Dose	-	-	+	-
Wier	-		+	+
Min/Max validated UVT	-		+	-
Flow	-		+	-
DPM	-		+	-
Channel configuration	-		+	V
UVT – LF factor	-		+	-

6.2 Controller Screens

This section describes the different screens of the Controller.

6.2.1 Main Screen

The Main screen displays a graphical representation of the OCUVS status.

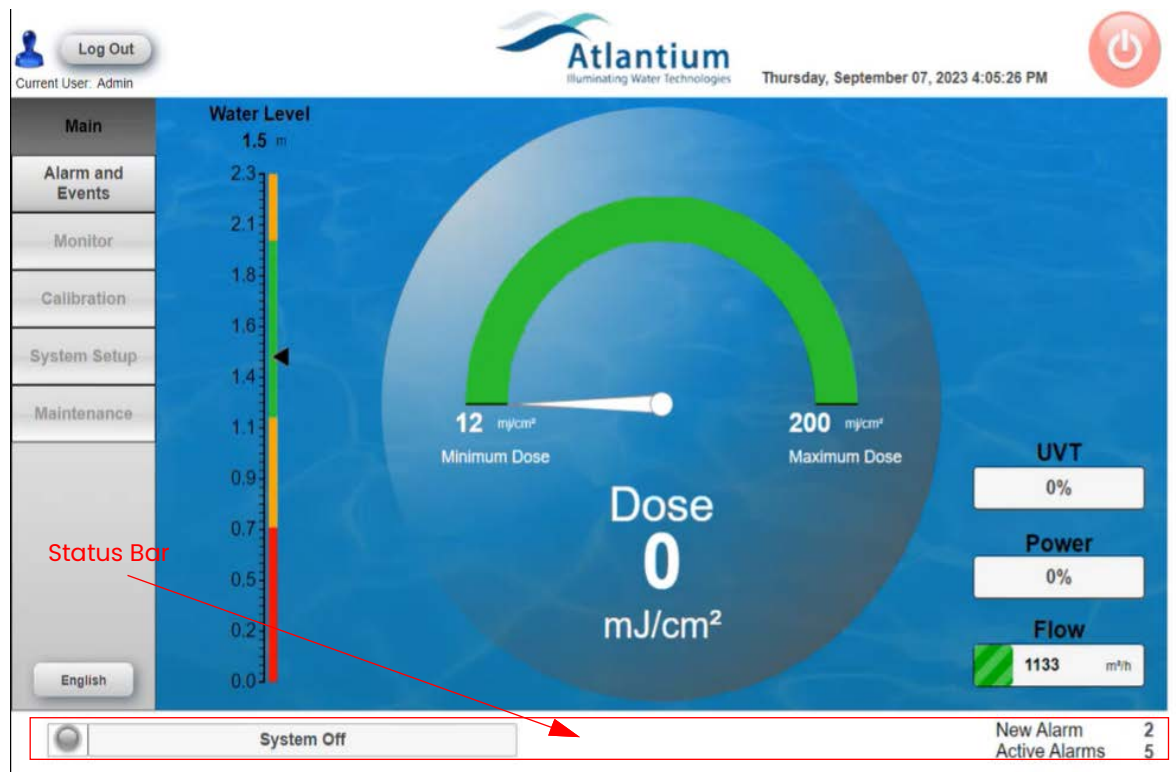


Figure 6-2: Main Screen

The system Main screen shows the dose or power output, depending on system settings (See **System Setup Screen** on page 70.)

Dose mode reading is in mJ/cm^2 . Dose value is derived from readings from the UVIS sensors, which measure the direct output from the lamps, and readings from the UVT sensors, which measure lamp output transmitted through the water.

The UVT sensor measures the water transparency in [mA] units in the range 4-20mA $\pm 0.2\text{mA}$ where 4mA $\pm 0.2\text{mA}$ is UVT=0.

Power mode status is S% and is a measure of the power supplied directly to the lamps.

All lamps will illuminate at the same intensity and lamp efficiency is similar for all lamps in a module Bank/Cassette

1. Current status of calculated in percentages:

$$S = \frac{UVIS}{UVIS_{T0}} = \frac{UVIS[\text{mA}] - UVIS_{bias}[\text{mA}]}{UVIS_{T0}[\text{mA}] - UVIS_{bias}[\text{mA}]}$$

Where:

UVIS is a value of the lamp UV-intensity sensor signal in [mA]

$UVIS_{bias}$ is a value of UVIS sensor bias (zero radiance. Shall be about 4[mA])

$UVIS_{T0}$ is a value of the UV-intensity sensor signal in [mA] during the calibration.

The Water level reading is derived from the water level sensor that is installed close to the first cassette of lamps in each channel.

Status Bar - Displays current system's status

6.2.2 Alarms & Events Screen

The Alarms and Events screen includes three tabs:

- Alarms - displays the latest alarms
- Alarm History - Alarms Log
- Events - events that occurred during operation

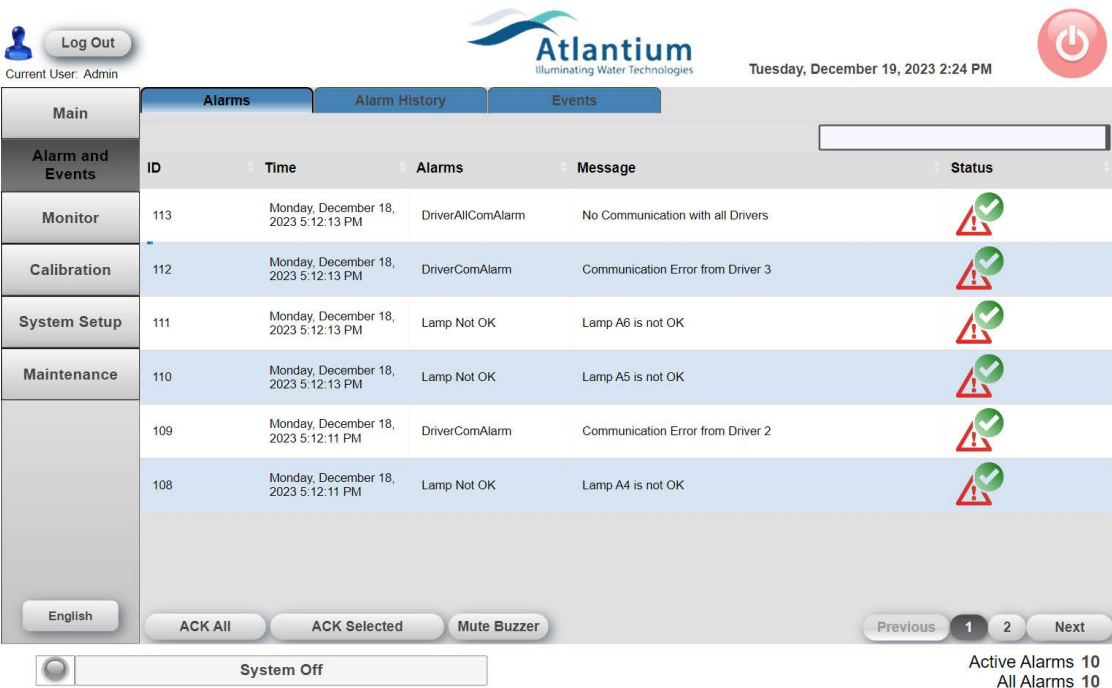


Figure 6-3: Alarm Screen

The following table lists different Alert messages, possible alarm triggers and error severity.

Table 6-2: Alerts

Alert name	Alert text	Trigger	Delay/Timeout	Severity
Minimum Water Level	Water level is below the Minimum Level	Water Level < Minimum Water level	Time delay for Action Minimum Water level Notice: in case we return to valid water level timer will be initialize and warning won't invoke	Non-Critical
Minimum Water Level -Shutdown	Due to Water Level below the Minimum – The System will Shut Down	Water Level < Critical minimum water level	Time delay for Critical minimum water level Action Notice: in case we return to valid water level timer will be initialize and warning won't invoke	Critical
Top Water Level	Max level of water is above Max Water level	Water Level > Maximum Water level	Time delay for Action Maximum Water level Notice: in case we return to valid water level timer will be initialize and warning won't invoke	Non-Critical
DPM Time out	DPM did not reach to the top/bottom position At Bank/Cassette ID #		DPM Max Cycle time	Non-Critical
DPM Maintenance	DPM Maintenance is required	Next Maintenance date >= current date		Non-Critical Once Next Maintenance date will be changed – Alarm will be cleared
Dose off spec.	System is Off-Spec, while the UV Dose Value is below the defined Minimum Dose	Minimum Validated Dose>Actual Dose	Off Spec.delay	Non-Critical
UVT off spec.	System is Off-Spec, since the UVT value is below Minimum Defined UVT	Measured UVT<Minimum UVT	Off Spec.delay	Non-Critical
Target Dose	System UV Dose is below the Set-Point UV Dose	Actual Dose< Target Dose	Off Spec.delay	Non-Critical

Table 6-2: Alerts

Alert name	Alert text	Trigger	Delay/Timeout	Severity
Lost Communication	Lost Communication - Modbus- System Fail ■ Lost communication with 1800 Card # ■ Lost communication with Driver Cabinet # ■ Lost communication with Flow Meter (in case flow is transmitted via communication)	In case Controller fails to Receive/ Transfer data Action: Shutdown	Communication Time out	Critical
UVT/ UVI Sensor Fault	UVT/UVI Sensor is disconnected	UVT/UVI Sensor reading = 0 [mA]	Communication Time out	Critical
Flow/Weir/Water Level Sensor Fault	Flow/Weir/Water Level Sensor disconnected	Flow/Weir/Water Level Sensor reading = 0 [mA]	Communication Time out	Non-Critical
Lamp not OK	Lamp xx	Driver indication of Lamp Error/ Warning	Driver Time out defined at Driver	Non-Critical
System Shutdown	System Shutdown due to Lamps failure Bank# : xx Lamp ID:xx	Register 16-b11 =0 Or Register 16-b12 =0		Critical: (Banks with UVI/UVT Sensors) At Bank with 11 Lamps: Lamp ID 2 At Bank with 6 Lamps: Lamp ID 1 Non-Critical All other Lamps

- Active Alarm (Critical/Non Critical) - turns on red/ Green light indicators on the cabinet door.
- On HMI:
 - New Alarm - increments new Alarm counter.
 - Active Alarm - increments Active Alarm counter.
- Each New Alarm appears with a gray check  icon
- Pressing Acknowledge turns the check icon  to green
- Unless Alarm is Inactive - light indicator on cabinet will stay ON

6.2.3 Alarm History View

The following screen logs all the Alarms that occurred during the OCUVS activity.

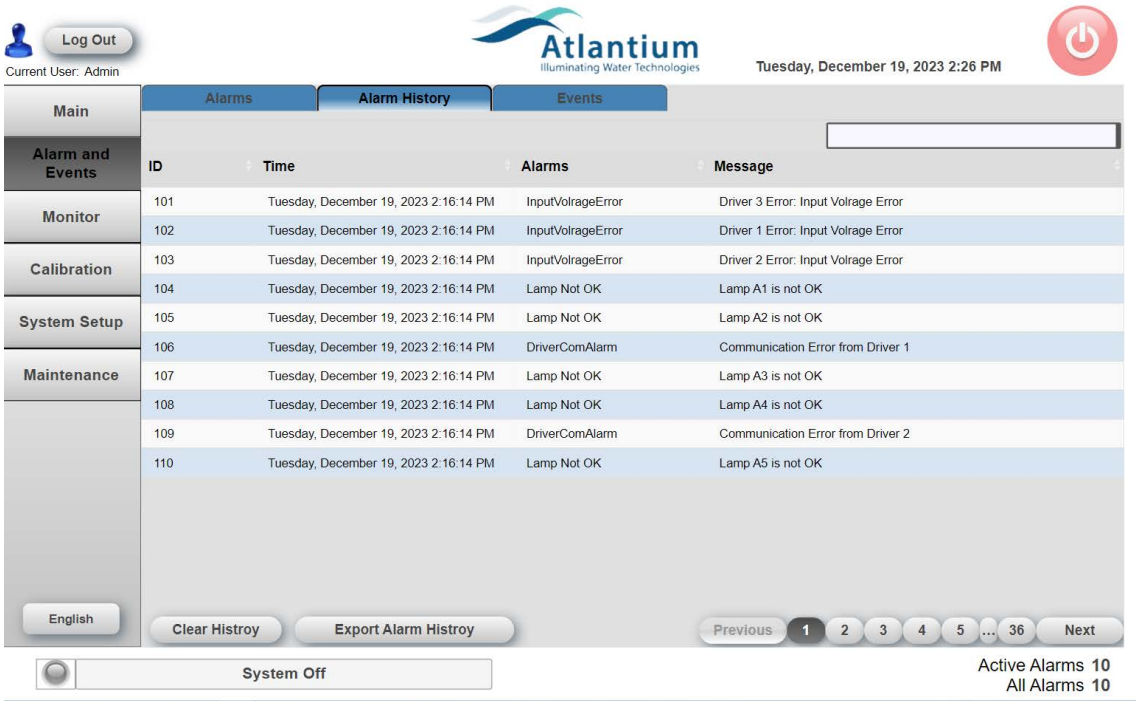


Figure 6-4: Alarm History Screen

The Alartm History screen includes the following details:

- ID – Alarm’s ID
- Time – time of the alarm
- Alarms – Alarm’s description
- Message – error’s short description

6.2.4 Events View

The Events Screen lists the event that occurred during the OCUVS activity.

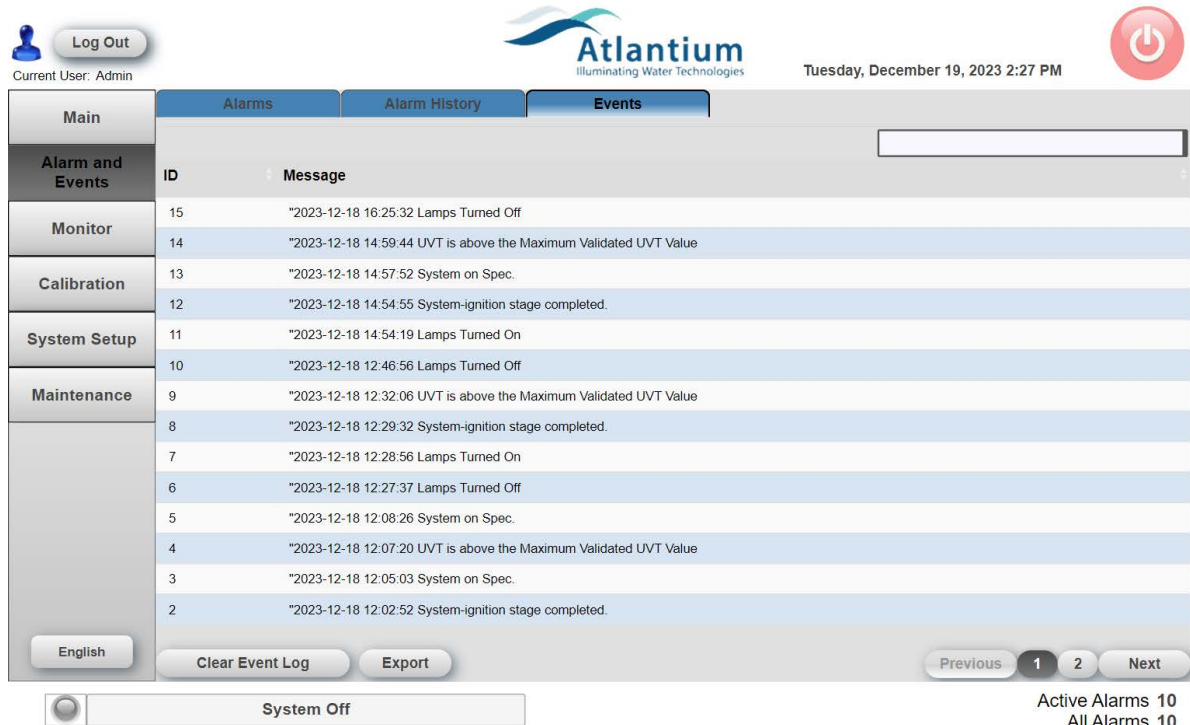


Figure 6-5: Events Screen

The Events screen includes the following details:

- ID - Events ID
- Messages - description of the event

Each Event name appears on bottom line and is recorded under Alarm & Events/Event.

Table 6-3: Events

Event name	Event text	Trigger
Controller turned On	Power was restored to the control module.	Turn ON electricity of controller
System turned Off manually	One or more Bank/Cassettes have been turned Off manually.	Turn OFF via HMI
System was turned On manually	One or more Bank/Cassettes were turned On manually.	Turn ON via HMI
System ignition stage completed	System-ignition stage completed.	ALL in use Bank finished ignition
System is On-spec	System is On-spec	ALL condition for On spec. are completed
Sensor Calibration completed	Flow Meter/Water Level / Weir Opening UVIS / UVTS Calibration completed	Sensor calibration Ended

6.2.5 Monitor Screen – Channel View

The position of the Weir gate (aka sluice gate) is monitored /controlled on this screen Figure 5-3 on page 44.

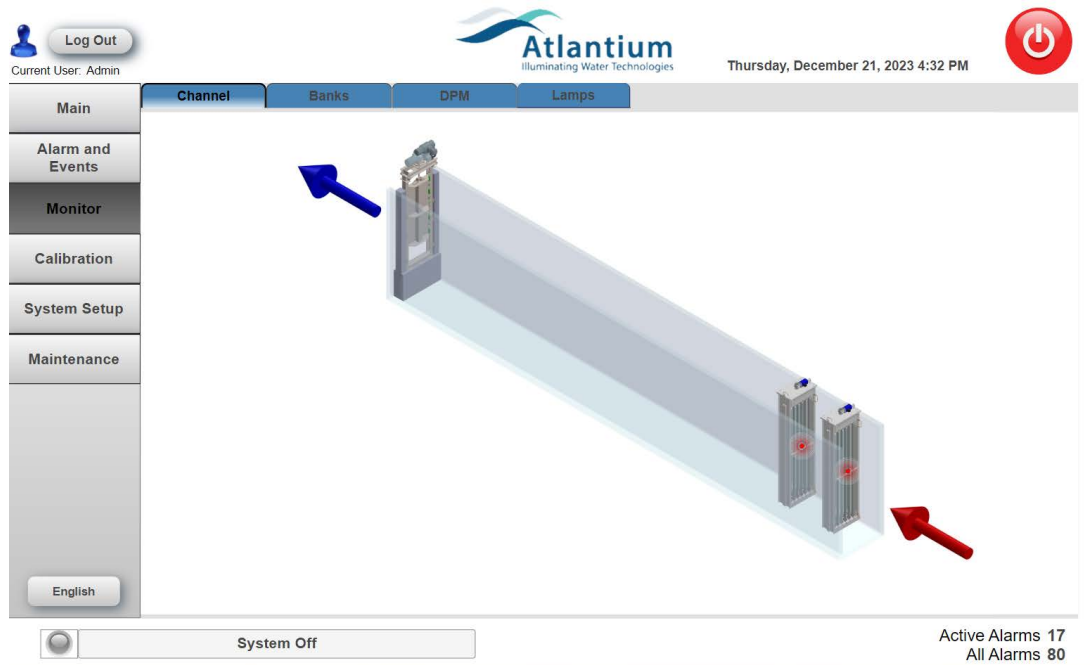


Figure 6-6: Channel View

The screen also shows the lamp cassettes/banks in the channel. Clicking on a bank (on the red spot) will lead to an overview focusing on that bank.

- When the system is off / in ignition process - **red LED** display on the bank
- At the end of the ignition process - **green LED** display on the bank
- If there is a malfunction in one lamp from the bank - **orange LED** display on the bank and an image of the bank appears on the screen with the malfunctioning bulb highlighted.
- Click Banks tab to move to Banks view.

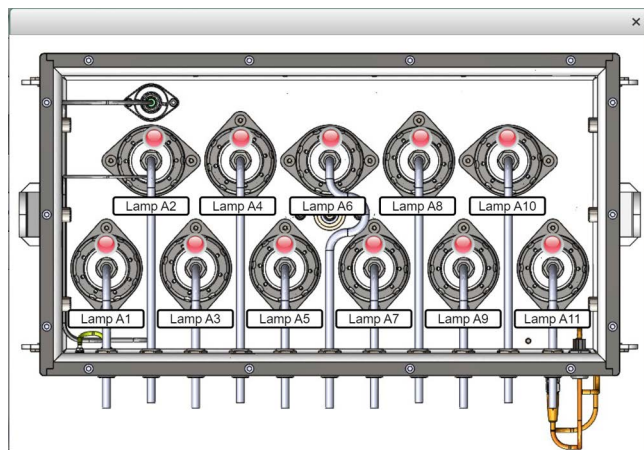


Figure 6-7: Bank Lamps View

6.2.6 Monitor Screen – Bank View

The functional state of the Banks (aka Cassettes) of UV lamps is displayed in the Banks view, see Figure 6-8, on page 62.

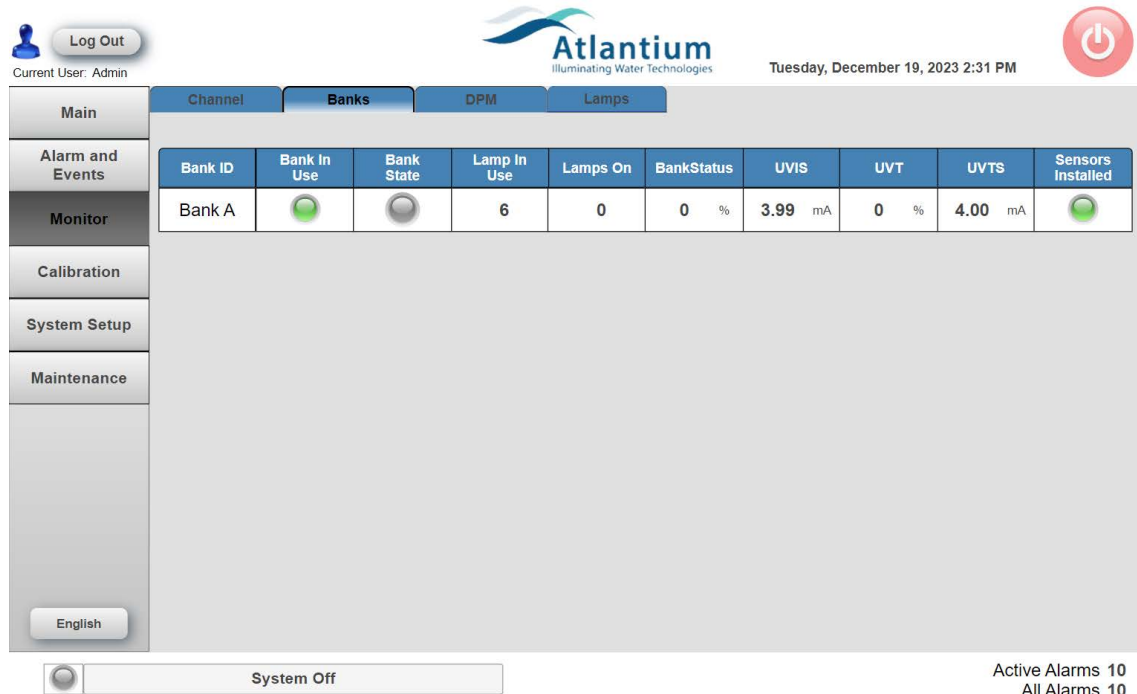


Figure 6-8: Banks/Cassettes view

On the Bank screen, a table will be displayed with a row for each bank present in the channel. The following data will be displayed in each row:

- Bank ID - Bank number. Clicking brings to a tabular tab of the specific bank (lamps)
- Bank In Use - Bank not in use
- Bank State - OFF/ON when the bank is in ON state and at least one lamp is lit
- Lamps in Use - number of lamps in use
- Lamps On - number of lamps ON
- Bank Status[%]- lamp status
- UVIS - current [mA] from the UVI sensor
- UVT - calculated UVT [%-1cm] at the bank
- UVTS - current [mA] from the UVT sensor
- Sensors installed- Yes/No

Click the DPM tab to move to DPM view, Figure 6-9, on page 63.

6.2.7 Monitor Screen – DPM view

This screen is used to switch the DPM on/off.

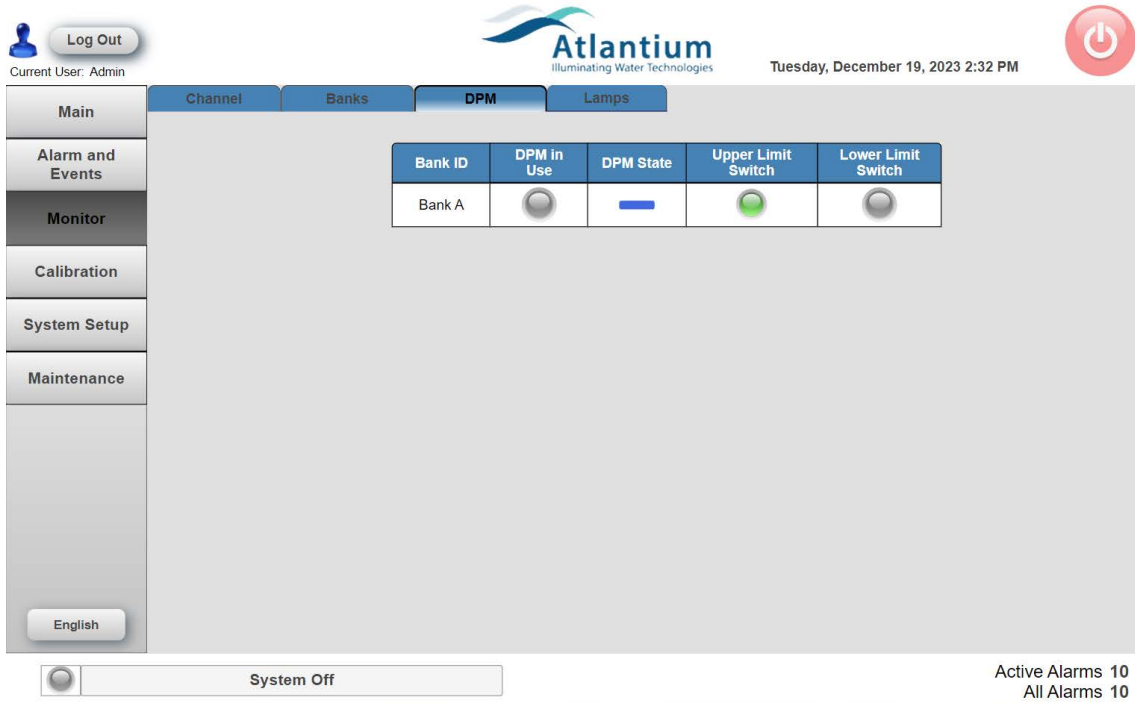


Figure 6-9: DPM View

- Bank ID – bank number.
- Direction – Up/Down/Idle
- DPM In use – On/Off
- Cycle Counter – the number of cycles performed by the DPM
- Upper LS – upper limit switch status
- Lower LS – lower limit switch status

Click Lamps tab to move to Lamps view, Figure 6-10, on page 64.

6.2.8 Monitor Screen – Lamps View

On the Lamp View screen you can select the Bank to monitor, and view lamps' specifications.

Log Out
Current User: Admin

Atlantium
Illuminating Water Technologies

Tuesday, December 19, 2023 2:33 PM

Channel Banks DPM Lamps

Select Bank to Monitor: A

Lamp	Power	Power	Voltage	Current
Lamp A1	0.0 %	0 W	0 V	0 mA
Lamp A2	0.0 %	0 W	0 V	0 mA
Lamp A3	0.0 %	0 W	0 V	0 mA
Lamp A4	0.0 %	0 W	0 V	0 mA
Lamp A5	0.0 %	0 W	0 V	0 mA
Lamp A6	0.0 %	0 W	0 V	0 mA

English

System Off

Active Alarms 10
All Alarms 10

Figure 6-10: Lamps view

At the top of the screen, select the Bank by Capital Letter (Drop Down List).

Displayed data:

- Lamp [ID] – the ID number of the lamp
- Power [%] – current working power of the lamp
- Power [W] – the lamp output
- Voltage [V] – lamp voltage
- Current [A] – lamp current

6.3 Calibration screen

The Calibration screen consists of two tabs:

- System Sensors
- Bank Sensors



When the Weir Type is set to Manual or Automatic in the System Setup tab screen, the Weir Height Sensor Calibration section appears on the System Sensors tab screen.

Setting the Flow Sensor Type to Flow Switch in the System Setup tab screen, displays the Flow Switch section instead of the Flow Sensor Calibration section.

6.3.1 System Sensors View

The System Sensors screen is used to calibrate the Flow Sensor, the Weir Height Sensor and the Water Height Sensor.

Log Out
Current User: Admin

Atlantium
Illuminating Water Technologies

Thursday, December 28, 2023 1:02 PM

Main
Alarm and Events
Monitor
Calibration
System Setup
Maintenance

System Sensors Bank Sensors

Flow Sensor Calibration

Point	Flow	Sensor Current
First	0 m ³ /h	4.00 mA
Second	300 m ³ /h	20.00 mA

Set Calibration

Flow Sensor Live Value

Flow	Sensor Current
48 m ³ /h	6.57 mA

Water Height Sensor Calibration

Point	Height	Sensor Current
First	2.0 m	4.00 mA
Second	0.5 m	20.00 mA

Set Calibration

Water Height Sensor Live Value

Height	Sensor Current
1.4 m	10.37 mA

English

System Off

Active Alarms 36
All Alarms 267

Figure 6-11: System Sensors Screen (Weir Type - None)

The System Sensors screen with no Wier gate used, includes the following sections:

Flow Sensor Calibration

- Point
- Flow
- Sensor Current
- Set Calibration

Flow Sensor Live Value

- Flow
- Sensor Current

Water Height Sensor calibration

- Point
- Height
- Sensor Current
- Set Calibration

Water height Sensor Live Value

- Height
- Sensor current

When a Weir gate is utilized (Manual/Automatic), the following section also appears:

Weir high Sensor Calibration

- Point -
- Opening -
- Sensor Current -

Log Out
Current User: Admin

Atlantium
Illuminating Water Technologies

Thursday, December 28, 2023 2:39 PM

Main
Alarm and Events
Monitor
Calibration
System Setup
Maintenance

System Sensors Bank Sensors

Point	Flow	Sensor Current
First	0 m ³ /h	4.00 mA
Second	300 m ³ /h	20.00 mA

Set Calibration

Flow	Sensor Current
48 m ³ /h	6.57 mA

Point	Opening	Sensor Current
First	0 %	4.00 mA
Second	100 %	20.00 mA

Set Calibration

Opening	Sensor Current
1 %	4.00 mA

Point	Height	Sensor Current
First	2.0 m	4.00 mA
Second	0.5 m	20.00 mA

Set Calibration

Height	Sensor Current
1.4 m	10.38 mA

English

System Off

Active Alarms 36
All Alarms267

Figure 6-12: System Sensors Screen (Weir Type - Automatic/ Manual)



Calibration of sensors is to be performed after system installation by installation team. Each sensor type has its particular calibration settings.

6.3.1.1 Flow Sensor Calibration

Calibration of the Flow sensor has two options:

- Flow Meter
- Flow Switch

Flow Meter/Switch Calibration

- If a Flow meter is installed, it has an output of 4-20 mA. 4 mA is defined as 0 flow, 20 mA = max flow
- If a Flow switch is installed the estimated flow is set to be displayed when water flow is detected

The calibration procedure is carried out by an Atlantium authorized technician who will test the system in specific states, for example:

- Step 1.** Open the weir (sluice gate) to achieve a specific desired flow rate.
- Step 2.** Measure the actual flow rate obtained and the sensor reading in mA
- Step 3.** Record these 2 values and repeat for different flow rates
- Step 4.** The system will set up a function of flow rate against sensor reading, to accurately measure flow rate for the control of the system.

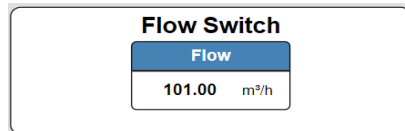


Figure 6-13: Flow Switch Sensor Type

6.3.1.2 Water Height Sensor Calibration

If a Water Level sensor switch is installed to control opening /closing of the weir, then Weir Height Sensor Calibration & Weir Height Control window with Manual/Automatic modes will appear on screen.

- Step 1.** The measurement from this sensor is calibrated by measuring actual water level and noting the corresponding sensor output current in mA.
- Step 2.** Repeat Step 1 above for a different water level.
- Step 3.** Enter the calibration readings. The system will create a function for reliable water level readings to be received at controller.

6.3.2 Bank Sensors View

The Bank Sensors screen is used to calibrate the UVI and the UVT sensors.



The UVI and UVT Sensors can be calibrated separately or both at the same time.

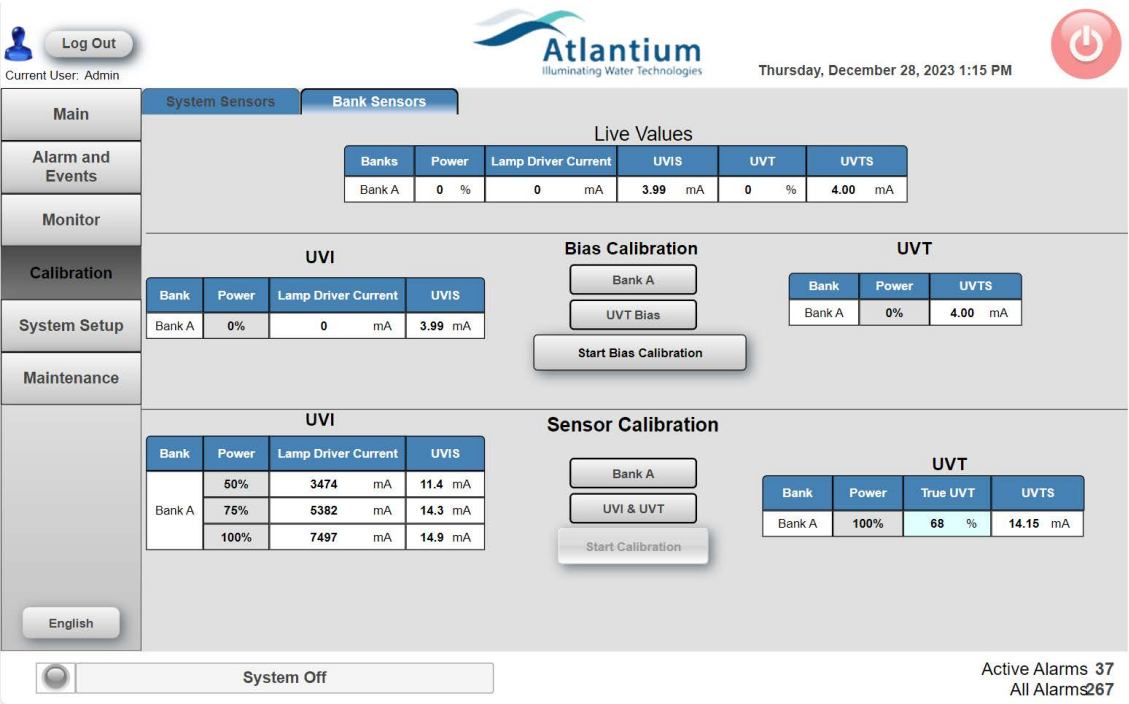
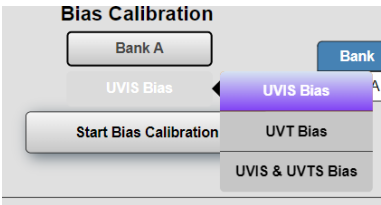


Figure 6-14: Bank Sensors Screen

6.3.2.1 Bias Calibration

- Step 1.** Turn off the OCUV system and wait for two minutes
- Step 2.** Select the Bank of which sensors you would like to calibrate



- Step 3.** Select the sensor or both sensors to calibrate.
- Step 4.** If the DPM is active, the following message will be displayed:
Calibration will commence once DPM cycle concludes. This may take several minutes.

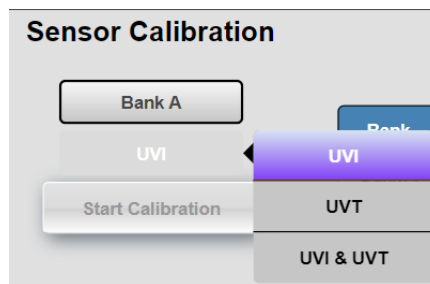
est

Step 5. Click Start Sensor Bias Calibration.

6.3.2.2 Sensor Calibration

➔ UVI Calibration

- Step 1.** Click on Start, the lamp in the specific Cassette/bank with a UVI connected will be turned on at 100%.
- Step 2.** Wait until all Banks have completed the ignition phase.
- Step 3.** Allow the lamps to lit for 20 min before commencing calibration.
- Step 4.** Select the Bank of which sensors you would like to calibrate
- Step 5.** Select the sensor or both sensors to calibrate.



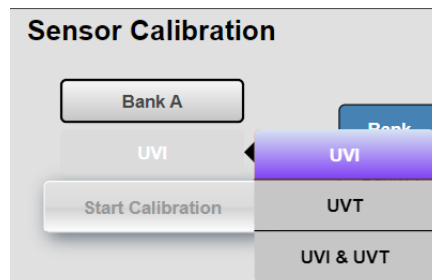
- Step 6.** After verifying that the lamp is 100% lit and stable, press the Start Calibration button (or alternatively the Stop button to stop the process).
- Step 7.** The system will activate all lamps for the given calibration time (min) at a power of 100%, and record the output current from the UVI sensor and the lamp driver.
- Step 8.** Repeat **Step 7.** at 75%, 50% power and after half a minute between each transition, it will collect the current value from the sensor and the driver.

The following function will be used for lamps that do not have a UVI sensor.

- Step 9.** The system will use these points to build a function of the form $y(x)=a \cdot x+b$, to convert the current value of the driver to the current value of the sensor.

➔ UVT Calibration

- Step 1.** Click on Start, the lamp in the specific Cassette/bank with a UVI connected will be turned on at 100%.
- Step 2.** Wait until all Banks have completed the ignition phase.
- Step 3.** Allow the lamps to lit for 20 min before commencing calibration.
- Step 4.** Select the Bank of which sensors you would like to calibrate.
- Step 5.** Select the sensor or both sensors to calibrate.



- Step 6.** After verifying that the lamp is 100% lit and stable, press the Start Calibration button (or alternatively the Stop button to stop the process).
- Step 7.** The system will activate all lamps for the 7 min at a power of 100%, and record the output current from the UVI sensor and the lamp driver.

6.4 System Setup Screen

The System Setup screen includes settings to be set by Atlantium authorized technician. The screen also includes settings that can be set by the operators.

System Setup Screen

Log Out
Current User: Admin

Atlantium
Illuminating Water Technologies

Monday, December 18, 2023 12:40 PM

Save Parameters

Channel Configuration	Flow Parameters	Water Level Parameters
Number of Banks in Channel: 1	Flow Sensor Type: Flow Meter	Parameter Value Action Time
Number of Lamps Per Bank: 6	Minimum Validated Flow: 20 m³/h	Maximum: 1.8 m Alarms 0.1 min
Number of Columns: 1	Maximum Validated Flow: 300 m³/h	Target Height: 0.0 m
Number of Rows: 1	Flow Units: m³/h	Minimum: 0.9 m Alarms 1.1 min
		Critical Minimum: 0.5 m Alarms 1.0 min
		Water Level Units: Meters

Process Parameters	DPM Parameters
Process Mode: Power Mode	Cleaning Pass Timeout: 301 s
Power Target: 100 %	Time Between Cycles: 0.5 min
	Time Until Maintenance: 1.00 y

Time Parameters	UVT Parameters
Communication Timeout: 30 s	Minimum Validated UVT: 20 %
Alarm Delay: 30 s	Maximum Validated UVT: 75 %

Dose Parameters	Driver Cabinet Parameters
Minimum Validated Dose: 11 mJ/cm²	Fan Stop Delay: 5 s
Maximum Validated Dose: 9,999 mJ/cm²	Dose Factor: 1.00
	LampType: 48
	Calibration Time: 2 min

English

Lamps On - On Spec

Active Alarms 1
All Alarms 1

Figure 6-15: System Setup Screen

6.4.1 Parameter Settings

For further information on the System Setup screens, please refer to Section 5.3 on page 46.

7 Open Channel UV System Maintenance & Troubleshooting

7.1 Maintenance Screen

The Maintenance screen includes maintenance procedures and displays different status parameters of the system.

7.1.1 Lamps View

The screenshot shows the 'Lamps' tab selected in the Atlantium interface. The top navigation bar includes 'Log Out', 'Current User: Admin', the Atlantium logo, the date 'Tuesday, December 19, 2023 3:12 PM', and a 'Save Parameters' button. The left sidebar contains menu items: Main, Alarm and Events, Monitor, Calibration, System Setup, and Maintenance (which is highlighted). The main content area displays a table of lamp data and a 'Select Bank' dropdown set to 'A'.

Lamp	Age	Ignition Count		Off/On	Power
Lamp A1	1 h	21	Replace	☐	50 %
Lamp A2	1 h	21	Replace	☐	50 %
Lamp A3	1 h	19	Replace	☐	50 %
Lamp A4	1 h	18	Replace	☐	50 %
Lamp A5	1 h	20	Replace	☐	50 %
Lamp A6	1 h	18	Replace	☐	50 %

At the bottom of the screen, there is a 'System Off' indicator and a status summary: 'Active Alarms 10' and 'All Alarms 10'.

Figure 7-1: Lamps Screen

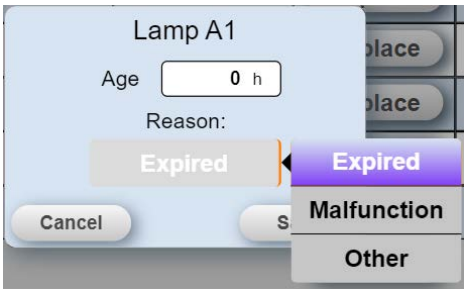
After you have replaced a UV lamp, fill in the following details:

Step 1. Click Replace to enter the new Lamp's age (hours of usage).

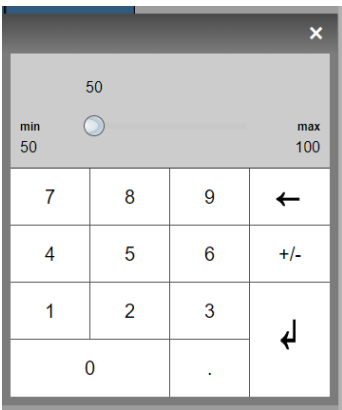
A modal dialog box titled 'Lamp A1' is shown. It contains an 'Age' field with a value of '0 h'. Below it is a 'Reason:' label followed by a button labeled 'Expired'. At the bottom of the dialog are 'Cancel' and 'Save' buttons.

Step 2. Click Replace to enter the new Lamp's Age (hours of usage).

Step 3. Click the Reason drop down list and select the reason why the lamp was replaced.



- Step 4.** Slide the ON/Off button to lit one or more lamps.
- Step 5.** Click on the Power % and set the lamp's power % .



When activating few lamps the power % should be the same.

Table 7-1: Lamps View Table

Parameter	Description
Lamp	Lamp's name
Age	Hours of usage
Ignition Count	Number of times the lamp was ignited
Replace/Install	Lamp's details
Off/On	Turn on/off a single or more lamps
Power	Lamp's power %

7.2 Drivers View

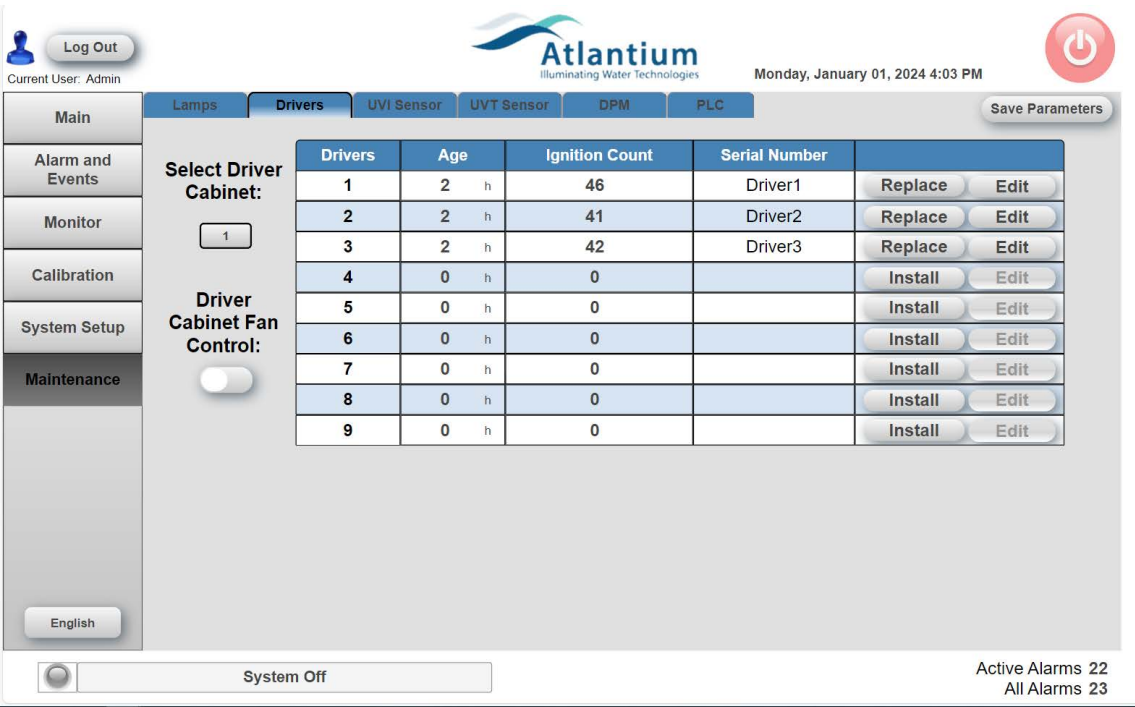


Figure 7-2: Drivers View Screen

Fill in the following details after replacing a driver:

- Step 1. Select the required driver cabinet.
- Step 2. Click Replace and enter the driver’s serial number.

Node 1

Serial Number

Reason:

Expired

Cancel

Save

- Step 3. Click the Reason drop-down list and select the reason for replacing the driver.
- Step 4. Click Edit to change the serial number (if needed).

Table 7-2: Drivers View

Parameter	Description
Drivers	Driver's name
Age	Usage time
Ignition Count	Number of ignition
Serial Number	Driver's serial number
Replace/Install	Maintenance action's details
Edit	Editing details

7.3 UVI Sensor View

The screenshot displays the Atlantium UVI Sensor View interface. The top navigation bar includes 'Lamps', 'Drivers', 'UVI Sensor' (selected), 'UVT Sensor', 'DPM', and 'PLC'. The sidebar on the left contains 'Main', 'Alarm and Events', 'Monitor', 'Calibration', 'System Setup', and 'Maintenance' (selected). The main content area shows a table with the following data:

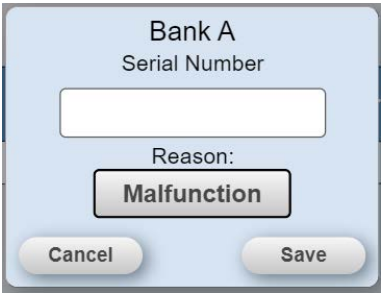
UVI	Age	Serial Number	
Bank A	618 h	UVI	Replace Edit

The bottom status bar indicates 'System Off' and shows 'Active Alarms 10' and 'All Alarms 10'.

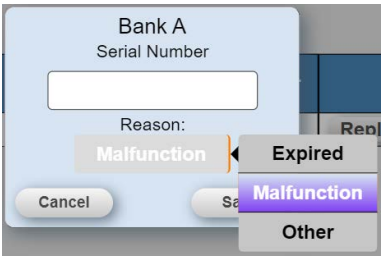
Figure 7-3: UVI Sensor

Fill in the following details after replacing a sensor:

Step 1. Click Replace and enter the sensor's serial number.



Step 2. Click Reason and select the reason for the maintenance procedure.



Step 3. Click Edit to change Serial number (if needed).

Table 7-3: UVI Sensor

Parameter	Description
Drivers	Driver’s name
Age	Hours of Usage
Ignition Count	Number of ignitions
Serial Number	Driver’s serial Number
Replace/Install	Driver’s details
Edit	Change Details

7.4 UVT Sensor View

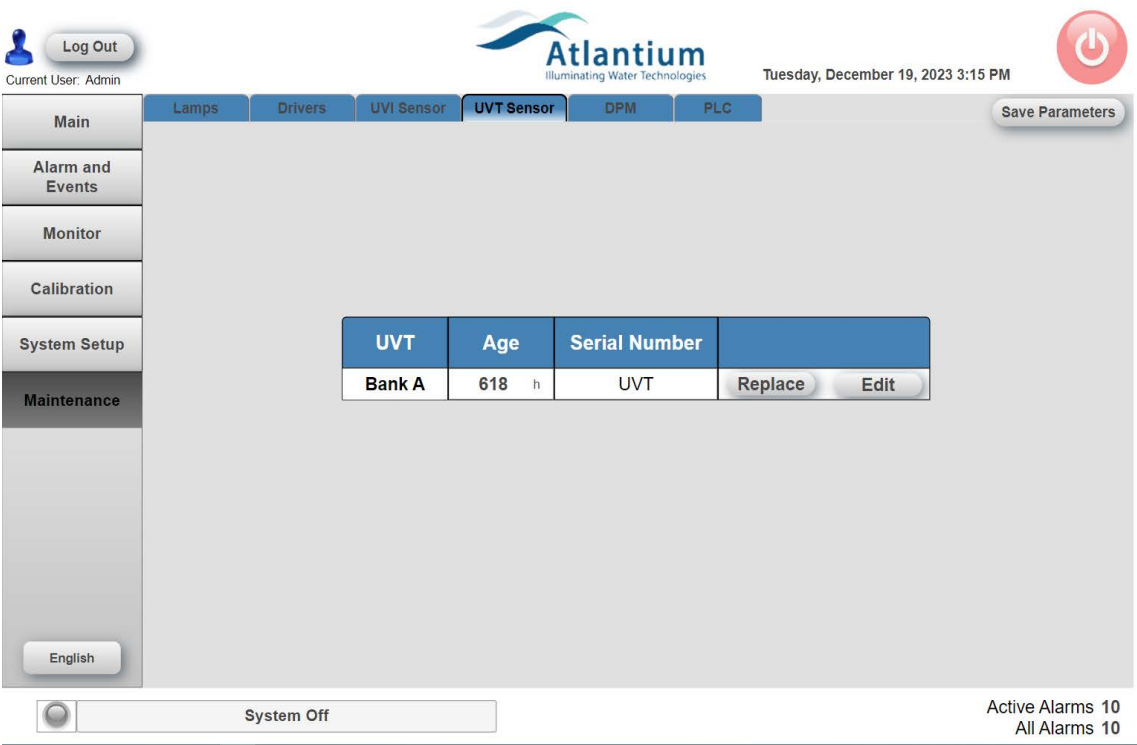
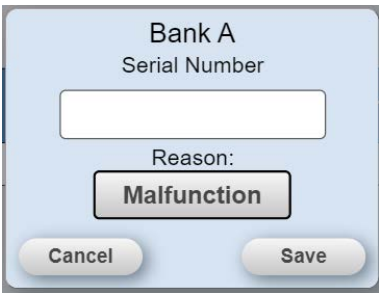


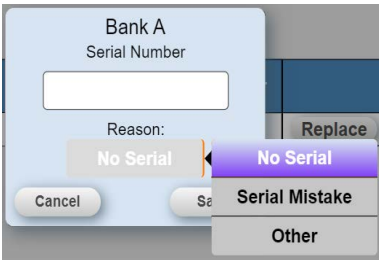
Figure 7-4: UVT Sensor View

Fill in the following details after replacing the UVT sensor:

Step 1. Click Replace and enter the sensor’s serial number.



Step 2. Click Reason and select the reason for the maintenance procedure.



Step 3. Click Edit to change Serial number (if needed).

Table 7-4: UVT Sensor

Parameter	Description
Drivers	Driver's name
Age	Hours of Usage
Ignition Count	Number of ignitions
Serial Number	Driver's serial Number
Replace/Install	Driver's details
Edit	Change Details

7.5 DPM View

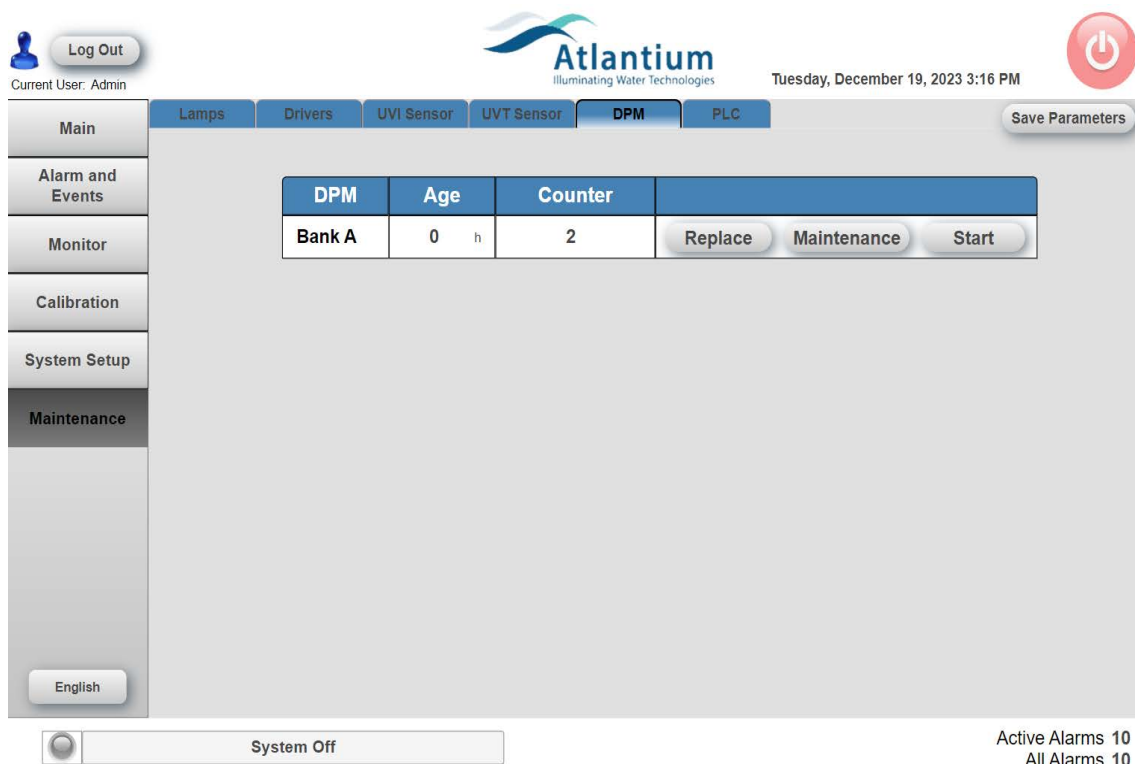


Figure 7-5: DPM Screen

After performing maintenance procedures for the DPM, fill in the following details:

Step 1. Click Replace and enter the Usage time in the Age field and the number of DPM cleaning cycles (Up /down) in the Count field.

Step 2. Click Reason and select from the drop-down list the reason for the maintenance procedure.

Bank A

Age

Count

Reason:

Expired

Step 3. Click Maintenance to lower the PDM assembly to the lowest point of the cleaning move. The Maintenance button text changes to Home.

Step 4. When maintenance is complete, click the Home button to revert the PDM assembly to its default position. The Home button text turns back to Maintenance .

Step 5. Click Save.

Table 7-5: DPM View

Parameter	Description
DPM	DPM Name
Age	Hours of usage
Counter	Number of cleaning cycles
Replace	DPM Details
Maintenance	Moves the DPM cleaning assembly down
Start	Initiates three movments (up/Down) cycle

7.6 PLC View

The PLC screen displays the PLC status. Table 7-6 describes the details displayed on the PLC View screen.

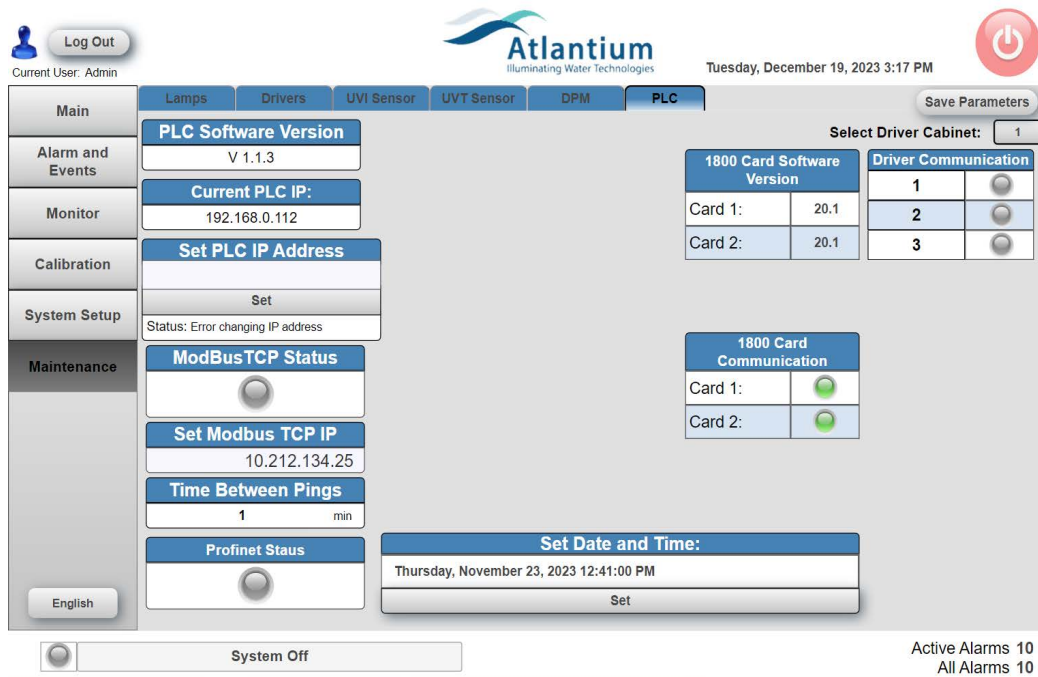


Figure 7-6: PLC Screen

Table 7-6: PLC View

Parameter	Description
PLC Software Version	PLC Software Version
Current PLC IP Address	Current PLC IP Address
ModBus TCP Status	ModBus TCP Status
Set ModBus TCP IP	Setting the Modbus TCP IP
Time Between Pings	Scada communication status(TCP IP protocol)
Profinet Status	Scada communication status(Profinet protocol)
1800 Card Communication	Communication status
Driver Communication	Communication status
Select Driver Cabinet	Select driver cabinet to check communication

7.7 Maintenance Procedures

The following procedures should be performed only by qualified personnel.

7.7.1 UVI/ UVT Replacement

Step 1. Unscrew the Socket Head cap screws (B).

Step 2. Remove the Cassette's top cover together with the DPM engine (A).

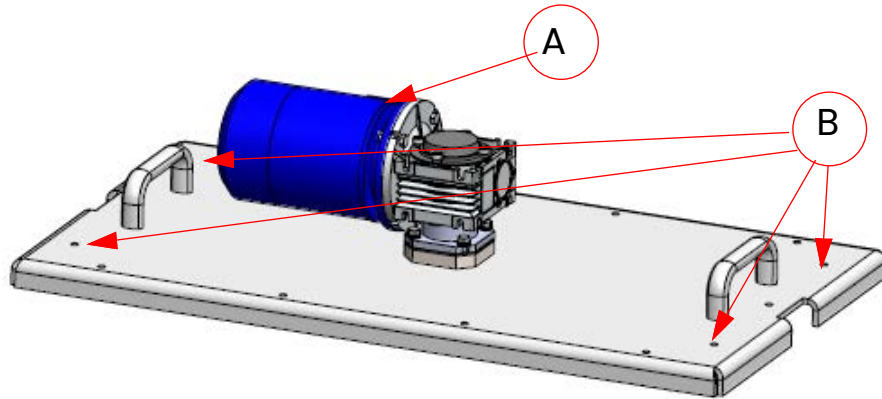


Figure 7-7: Top Cover and the DPM Engine

Step 3. Sensors replacement requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) for systems that have them, and the lamp be removed. In the section, See **Replacing a Lamp** on page 83. and remove the lamp connector assembly and lamp housing from both sides of the Unit, as well as the lamp, placing them in a secure location until they need to be reinstalled.

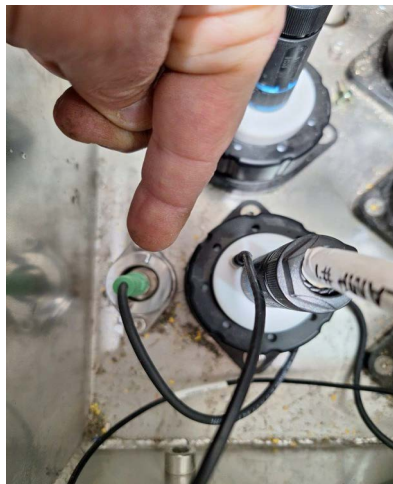


Figure 7-8: UVTS Connector

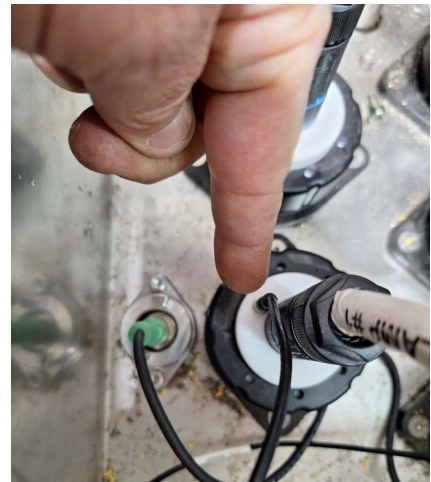


Figure 7-9: UVIS Connector

7.7.1.1 Replacing the UVIS

Step 1. Remove the Lamp Connector Housing.

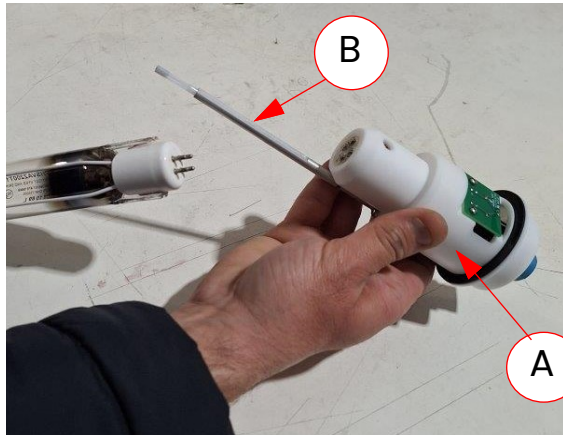


Figure 7-10: Removing the Lamp Connector Housing (A) and the Wave Guide (B)

Step 2. Place it on a stable surface.

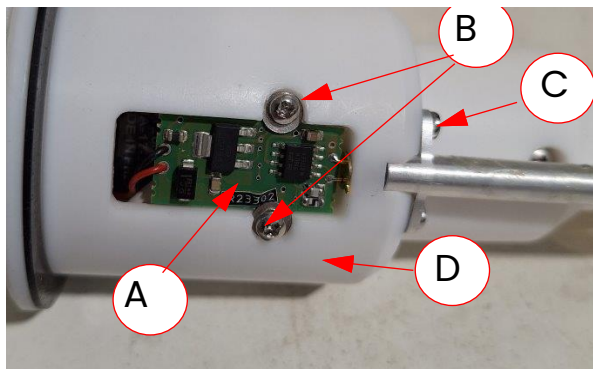


Figure 7-11: UVI Circuit (A)

Step 3. Unscrew the Wave Guide screws (C) and remove it from the Connector Housing (D).

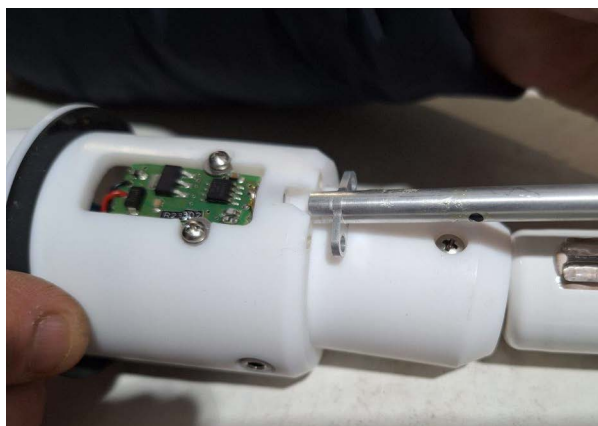


Figure 7-12: Removing the Wave Guide

Step 4. Release the Circuit screws.

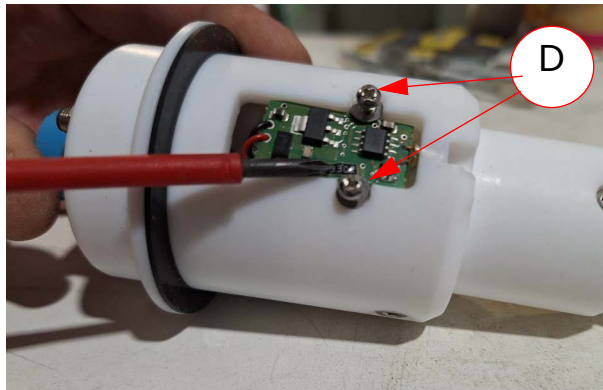


Figure 7-13: Removing the UVIS Circuit Screws (D).

Step 5. Carefully pull out the UVIS circulate and expose the UVI Sensor (A).

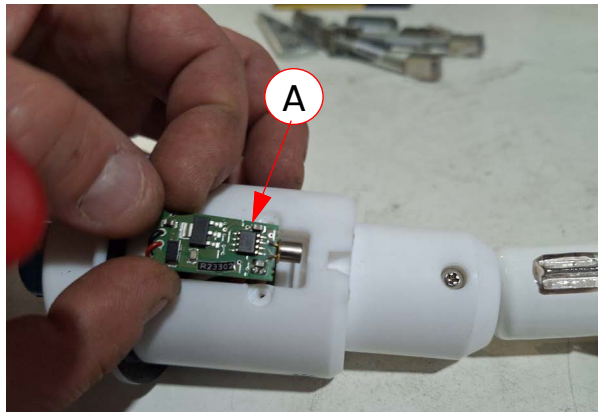


Figure 7-14: The UVI Sensor (A)

7.7.1.2 Replacing the UVTS

Step 1. Release the UVTS connector (F).

Step 2. Release the UVTS threaded nut (G)..

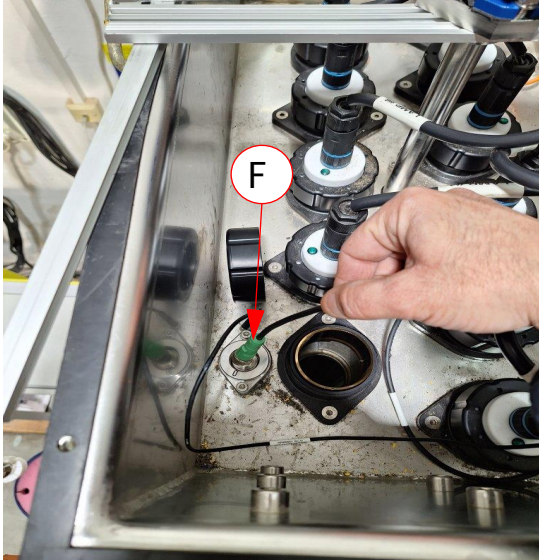


Figure 7-15: Releasing the UVTS Connector (F)



Figure 7-16: Releasing the UVTS Nut (G)

Step 3. Pull out the UVTS and remove it.

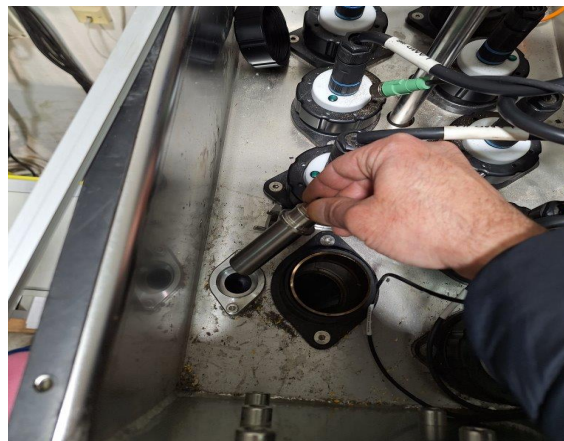


Figure 7-17: Removing the UVTS

7.7.2 Replacing a Lamp

When replacing a lamp please follow the following precautions:

Performed
as needed

The UV lamp can be used as long as it provides the required dose and efficient service. You should monitor a lamp when its service status (displayed on CM) has dropped below 80%. A lamp should be replaced **ONLY** when necessary.

The entire Atlantium system must be shut down before any lamp can be replaced.

- Before replacing a UV lamp, shut down the entire Atlantium system.
- The UV lamp can be used as long as it provides the required dose and efficient service. (Generally, a lamp service status of 80% or more is considered efficient service.) Each lamp's performance is measured directly by a dedicated UV Intensity (UVI) Sensor. A lamp should be replaced when its status or performance declines and the lamp no longer provides the required performance at the appropriate electrical cost
- Once a lamp passes its rated hours, be alert to any changes in efficiency that could indicate the need to change the lamp
- To alert operator to the decline in lamp output, set a minimum UVI level in the **Control Module**. A warning is sent to the operator at both the user-set level and the internally coded minimum.



UV Hazard

- Wear appropriate safety goggles with UV eye protection.
- Do not look directly into the UV light.



- Wear protective gloves while replacing a lamp.
- Wear protective eye-wear while replacing a lamp. Pay strict attention to the safety warnings and precautions in the **Safety Overview** on page 33 in the front of this manual.
- In case of lamp breakage, refer to the Safety instructions in **Safety Overview** on page 33.

ATTENTION!



- The **Open Channel UV System** cassette includes 6 or 11 lamps and each lamp has 1 connector at the top end.
- Each cable and lamp connector is labeled with number. When reconnecting, be sure that each connector is connected to a cable bearing the **SAME** number label.
- Before installing a new lamp, verify that it is clean and free of grease. If not, rub it gently with the provided lens cloth. Never touch the lamp with your bare hands – hold it only by the end-connectors. Oily residues from fingerprints can damage the lamp.

Prepare the following:

- Replacement lamp
- Lamp cleaning cloth (comes with the lamp)
- White cotton gloves
- Eye protection
- Caution signs

➔ To shut down the entire system:

- Step 1.** On the Main screen of the **Controller**, press . The Password dialog box appears.
- Step 2.** Enter the password (the default password is **1234**) and press **OK**. The Atlantium system is turned off.
- Step 3.** On the top-right of the screen, press . The button turns gray. Wait approximately three minutes for the **Ready Lamps Off** signal to transmit.
- Step 4.** Open the Electrical Cabinet and turn the main circuit breaker to the **Off** position.
- Step 5.** Open the Drivers Cabinet and turn the main circuit breaker to the **Off** position.
- Step 6.** Continue with the procedure, *To replace a lamp* below.

➔ To remove/replace lamp

- Step 1.** Before replacing a UV lamp, shut down the entire Atlantium system.
- Step 2.** Allow the lamps at least 10 minutes to cool down before starting.
- Step 3.** Remove the connector cable from the lamp.

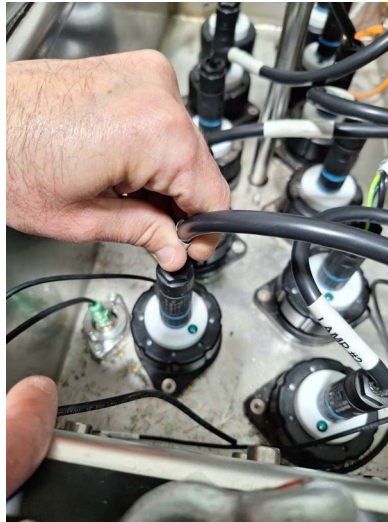


Figure 7-18: Removing the Lamp Connector Cable

- Step 4.** Disconnect the UVIS cable.
- Step 5.** Release and remove the Lamp Holder Nut.



Figure 7-19: Disconnecting the UVIS Cable

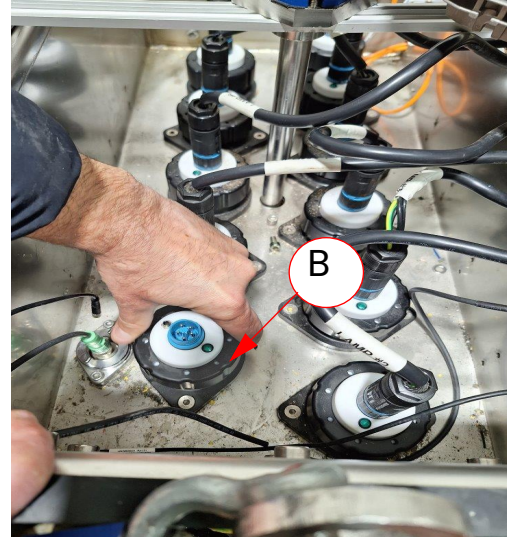


Figure 7-20: Releasing the Lamp Holder Nut (B)

- Step 6.** Wearing appropriate cotton gloves and skin covering, and using a suitable long-nose pliers, pull the lamp straight out of the quartz sleeve in the Unit and set it aside on a stable surface.
- Step 7.** To prevent any damage to the lamp, make sure you pull the lamp's connector assembly (A) straight out and not at an angle. Gently set the connector assembly aside on a stable surface.

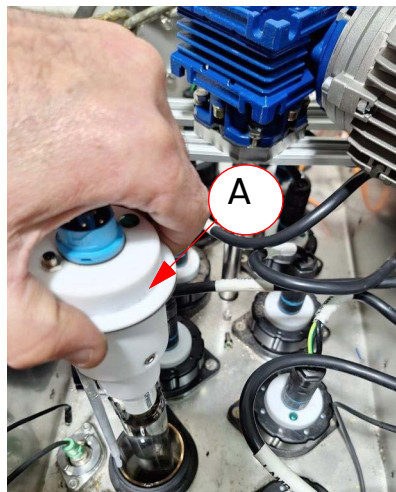


Figure 7-21: Pulling out the Lamp



Figure 7-22: Removing the Lamp Connector Housing

➔ **To Insert the new Lamp**

- Step 1.** Wearing appropriate cotton gloves and skin covering, clean the new lamp using the cleaning pad provided with the lamp. Verify that no visible fingerprint or dust is on the lamp..

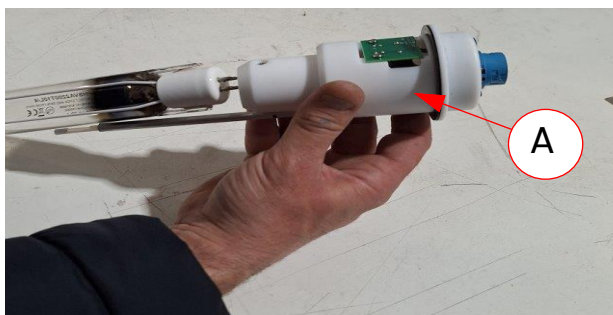


Figure 7-23: Inserting Lamp to the Connector Assembly (A)

- Step 2.** Connect the Bottom Lamp Adapter.

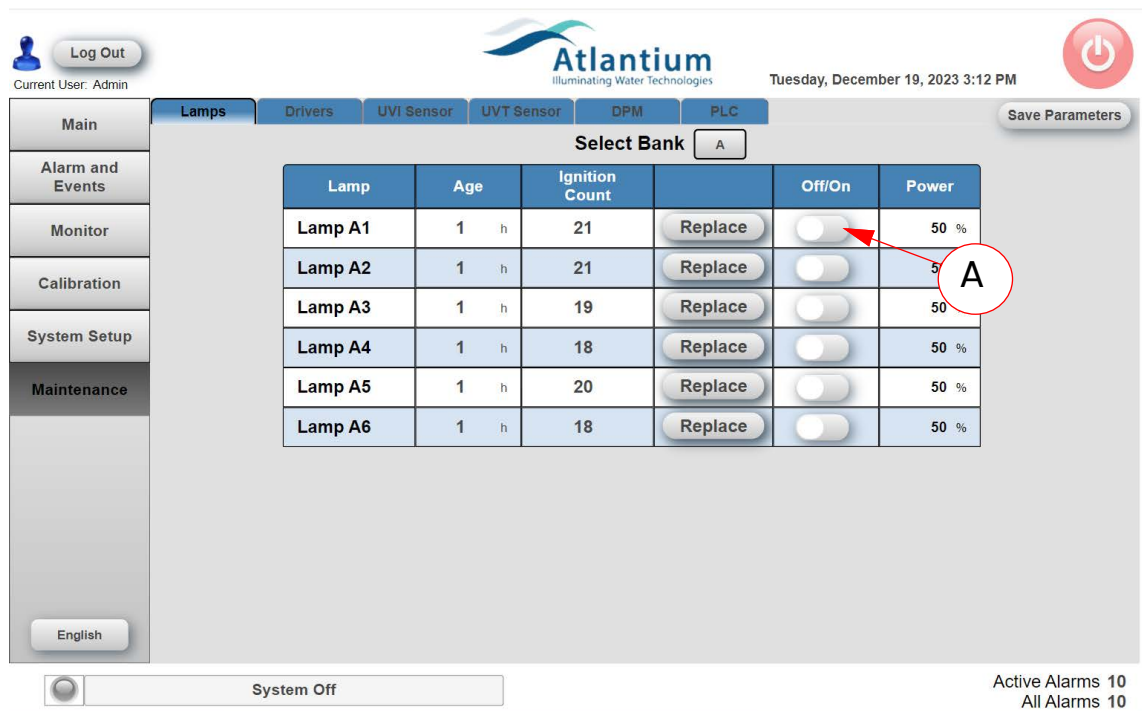


Figure 7-24: Connecting the Bottom Lamp Adaptor

- Step 3.** Carefully insert the new lamp into the quartz tube in the Unit, all the way until it stops.

➤ Reactivating the Lamp

- Step 1.** Open the Electrical Cabinet and turn the main circuit breaker to the **On** position.
- Step 2.** Open the Drivers Cabinet and turn the main circuit breaker to the **On** position.
- Step 3.** Log in the controller application.
- Step 4.** On the top-right of the screen, press . The button turns Green. Wait approximately three minutes for the lamps to lit.
- Step 5.** You can test and turn On specific lamp/s by switching the On/Off button (A) to **On**.



Log Out
Current User: Admin

Atlantium
Illuminating Water Technologies

Tuesday, December 19, 2023 3:12 PM

Save Parameters

Select Bank A

Lamp	Age	Ignition Count		Off/On	Power
Lamp A1	1 h	21	Replace		50 %
Lamp A2	1 h	21	Replace		50 %
Lamp A3	1 h	19	Replace		50 %
Lamp A4	1 h	18	Replace		50 %
Lamp A5	1 h	20	Replace		50 %
Lamp A6	1 h	18	Replace		50 %

English

System Off

Active Alarms 10
All Alarms 10

7.8 Cleaning/Replacing a Lamp's Quartz Sleeve and O-rings

The quartz sleeve is cleaned as needed according to the UVT Sensor readings and when deposit collects on the quartz sleeve. **The quartz sleeve's rubber O-rings must be replaced when dismantling the quartz sleeve.**

Under normal environmental conditions, the quartz sleeve's rubber O-rings must be replaced once a year even if they pass visual inspection. For environments where the water temperature is above 70°C (160°F), the quartz sleeve's O-rings must be replaced quarterly, for which after the first year, O-rings may require replacement less frequently. For more information, consult your Atlantium representative.

Depending on the quality of the water being treated by the Atlantium Unit, the exterior surface of the quartz sleeve that houses a lamp may require periodic cleaning or replacement if after the CIP process the quartz sleeve is still not sufficiently clean.

Prepare the following:

- Required O-rings
- #4 Allen wrench
- Sleeve insert assembly (Sleeve Tool)
- Protective gloves
- Appropriate cleaning cloth. A mild, household cleaning solution
- Alcohol



Wear protective gloves. To protect the replacement quartz sleeve, do not touch the internal surface with your bare hands.

➔ **To clean or replace the quartz sleeve:**

- Step 1.** Open each of the Electrical and Drivers cabinets and turn the main circuit breakers to the **Off** position.
- Step 2.** Quartz sleeve removal requires that the lamp connector assemblies, lamp housings (on both ends of the lamp) and the lamp be removed. Follow the steps in the section, ***Replacing a Lamp*** on page **83** and remove the lamp connector assembly and lamp housing from the side of the Unit with the sensor, as well as the lamp, placing them in a secure location until they need to be reinstalled.
- Step 3.** To prevent any damage to the sleeve, make sure you pull the sleeve straight out.



While pulling the lamp housing away, **DO NOT** tilt it and **DO NOT** pull it away from the lamp/sleeve axis at an angle.

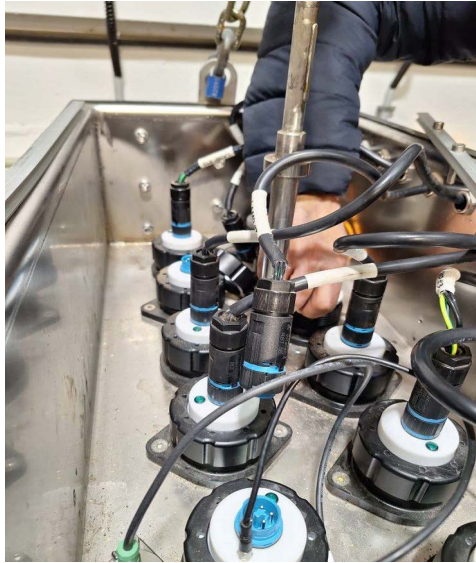


Figure 7-25: Pulling out the Quartz Sleeve



Figure 7-26: Sleeve Pulled Out

➔ Cleaning the Quartz Sleeve

- Step 4.** If there is a deposit buildup on the sleeve's exterior surface, clean it with a mild, household cleaning solution.
- Step 5.** Clean the exterior and interior surfaces with alcohol **(G)**. If the sleeve cannot be cleaned to a pristine condition, replace it with a new one.
- Step 6.** Ensure that there is no residual moisture or high humidity on the inner surface of the quartz sleeve. If any visible moisture exists, dry it before inserting it in place.

➔ Replacing the O-Ring

- Step 1.** Using a #4 Allen wrench, remove the two screws that secure the sleeve holder **(A)**.



Figure 7-27: Sleeve Holder assembly



Figure 7-28: Releasing the Sleeve Holder's Screws

- Step 2.** Remove the sleeve holder securing part.
- Step 3.** Hold the new rubber O-ring two parts and place them inside the O -ring rubber band (A).
- Step 4.** Place the rubber O-ring in its half part holder.

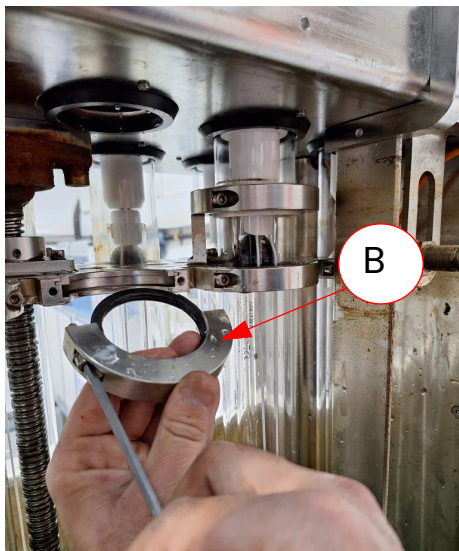


Figure 7-29: Removing the Sleeve Holder Securing Part (B)



Figure 7-30: Placing Rubber O-ring in rubber band (A)

Step 5. Insert the O-ring and the holder's half part into the PDM Sleeve Secure assembly.

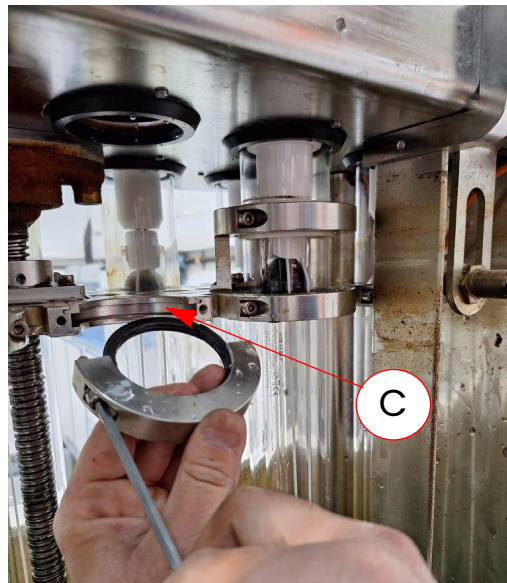


Figure 7-31: Placing the O-ring into the PDM Assembly (C)

Step 6. Using a #4 Allen wrench, fasten the screws of PDM assembly and tighten them.

➔ Returning the quartz sleeve

- Step 1.** Holding the sleeve, carefully insert the sleeve and make sure it moves smoothly into the Unit. Make sure it is centered.
- Step 2.** On the other side, guide the sleeve into place.

➔ Powering Up the System

- Step 1.** Open the Electrical Cabinet and turn the main circuit breaker to the **On** position.
- Step 2.** Open the Drivers Cabinet and turn the main circuit breaker to the **On** position.
- Step 3.** Log in the controller application.
- Step 4.** On the top-right of the screen, press . The button turns Green. Wait approximately three minutes for the lamps to lit.